

Book Notes & To-Do

A running scratchpad for the eventual book adaptation of this course

VIDEO GAME CREATION

How To Use This Document

This is a working scratchpad, not a polished outline - it exists so that adding a thought while the course is being built doesn't require deciding, right then, where it "really" belongs. Two rules keep it useful instead of becoming a mess:

- Add freely. A new To-Do or Note can sit next to, or even contradict, something written earlier - resolve the contradiction later, not in the moment. Rethinking the whole document every time something gets added defeats the point of having it.
- Reorganize rarely, on purpose. Periodic dedicated consolidation passes (see the standing To-Do item below) are where duplicates get merged, stale items get cut, and contradictions actually get resolved - not as a side effect of adding one new line.
- To-Do items get updated or deleted outright once actually resolved - this section should always reflect current, real open work, not a historical log. Notes & Commentary is the opposite: it's allowed to accumulate, and old entries are kept (even superseded ones) until a consolidation pass deliberately prunes them.

To-Do

Open items, pulled from the project's task list and from things noticed while building other documents. Check one off or delete it outright once it's actually done - don't leave resolved items sitting here.

- ☐ Book version: move the Godot signals sidebar ("Godot Signals: Create It, Connect It, Write the Function, Emit It", with the four-piece diagram_godot_signal_four_pieces) from the Lesson 8 Sidebars to Lesson 6, where signals first appear (the destroyed signal for hits), and remove it from Lesson 8. Lesson 8 should then only point back to it, since escaped is the second use of the same four pieces. For the current course the sidebar stays in the Lesson 8 Sidebars document, since no Lesson 6 Sidebars document exists yet. Source: topic in build_sidebars.js (lesson08 array). *(decision by Robert; book retrofit only)*
- ☐ Rule 15 (explicit, beginner-friendly names) retrofit: Godot's Label-node names were renamed this session (WinLabel/LoseLabel/ScoreLabel/LivesLabel/WaveLabel -> WinText/LoseText/ScoreText/LivesText/WaveText, matching GDevelop's existing text

object names) after a student read WinLabel as Window Label. Audit every other engine and lesson for names that lean on engine jargon (Label, Node, Actor, Pawn, Controller, Template, Trigger, Behavior, Instance) or that a newcomer could misread, rename them to plain names, and keep the same plain name in every engine per Rule 12. Update the Data Dictionary and Cross-Engine Consistency Reference to match. Also check Robert's own test projects, which still use the old names. *(Standards Rule 15, added per Robert's direction; Godot text-node names done in Lessons 6-14, nothing else audited yet)*

- ☐ Script/Visual parity cleanup for the book: for engines that offer both a scripting and a visual path (GameMaker GML Code vs. GML Visual, Unity C# vs. Visual Scripting, Unreal C++ vs. Blueprint), make sure the two paths produce the SAME gameplay and appearance - same positions, sizes, text, colors, timing, sounds, and variable names - so a student can switch paths without the game changing. Robert's own GameMaker Code and Visual projects have drifted apart through experimentation, and the drift is easy to repeat whenever a function call and a block do not map one-to-one or the options are not obvious. Do a side-by-side pass lesson by lesson: list every function call next to its equivalent block with identical field values, flag where an exact equivalent does not exist and say so explicitly (Rule 4/7), and re-check after every content change. Consider a short parity checklist per lesson in the Prep Checklist, and verify against Robert's actual test projects, not just the documents. *(raised by Robert; GameMaker is the known problem case)*
- ☐ Book retrofit naming decision (Robert): all asset/object names in the book use PascalCase with NO common prefixes or suffixes (no obj_, spr_, snd_, sfx_ and the like), e.g. GameManager, Enemy, PlayerShip. Assets are kept organized with each engine's own mechanisms instead - folders in GameMaker's and Unity's asset browsers, groups, Scenes, Content Browser folders, and so on - so related assets stay together without name prefixes. This applies to the book retrofit only: for the current course pass, existing variable-naming conventions for each engine stay as already established, and the shipped GameMaker content (obj_gamemanager and friends) is left alone. When the book retrofit happens, apply it in one pass across all engines, update Rule 12 and the Data Dictionary, and decide whether Lesson 2 setup should teach creating the asset folders. *(decision by Robert; supersedes the earlier separate asset-prefix and capitalization To-Dos)*
- ☐ Final deliverable for BOTH the course and the book: a Complete Script Document per engine and coding style - the entire finished Semester 1 project in one place (every object, script, event, function, graph, or block sequence in full, organized by object in build order, with the lesson that introduced each piece noted). Generate it from the same lesson build scripts so it cannot drift. Needed because lessons now show only the methods they add or modify (Standards Rule 14). SCOPE NOW FINALIZED as a live-course Lesson 15 deliverable (see the dedicated ToDo item below for the full decision and its explicit dependency on reviewing the remaining lessons first) - this entry now serves mainly as the original decision record; see the newer entry for the actual build plan. *(decision by Robert; Standards doc now lists it as a Final Deliverable; superseded in scope by the Lesson 15 Complete Codebase ToDo below)*
- ☐ Retrofit Standards Rule 14 (code completeness by level: Handout Level 1 complete code incl. initialization and display; Handout Levels 2-3 prose + diagrams; Teacher's Key complete code for all three levels) across Lessons 2-15 and all 8 engines. Lesson 8 GameMaker is the first case done (Handout Level 1 code box and Key Level 2/3 code). Known weak spot: most Keys describe Level 3 in talking-point prose only, and some Handouts show only the new lines. Audit lesson by lesson, starting with Lesson 8's other seven engines and Lesson 9. Scope note: Rule 14 requires COMPLETE code for every method a lesson adds or modifies (whole methods, not just the new lines), with the new

lines highlighted; unchanged methods are not repeated, since the Complete Script Document (separate To-Do) covers the whole project, because showing only new code has already confused students. Lessons 8-15 are brought up to this standard as each is reviewed; Lessons 2-7 are deliberately deferred to the book retrofit (they currently show new code only), and Lessons 16-30 must follow it as authored. *(Standards Rule 14, added per Robert's direction; only Lesson 8 GameMaker complete; Lessons 2-7 deferred to the book retrofit)*

- ☐ Finish the GameMaker __dnd_lives / __dnd_score review for Lessons 10-15 (restart/lives reset and any other lives/score use): replace deprecated built-in lives/score and "Set Instance Variable ... Lives/Score" with the purpose-built Lives/Score blocks (Target = obj_gamemanager) in Visual paths and __dnd_lives/__dnd_score in code paths. Deliberately held until Lessons 8 and 9 are fully reviewed. *(added after Lessons 8-9 GameMaker corrections; Lessons 10-15 not yet reviewed)*
- ☐ GameMaker: review every use of obj_name.variable notation in code paths and Visual paths. Where the script runs on the target object itself (or a block has a Target drop-down), prefer the plain variable name plus the Target drop-down over the dotted form, and make the code path and Visual path teach the same idea. Needs a deliberate pass across all lessons, not piecemeal edits. *(raised by Robert while reviewing Lessons 8-9)*
- ☐ GameMaker sidebar and comparison test: confirm that (a) the If Lives block with Target = GameManager, (b) an If Variable/Expression block testing __dnd_lives with Target = GameManager, and (c) a code if-statement using obj_gamemanager.__dnd_lives all behave identically, and write it up as a Sidebar (three ways to ask the same question about the same instance variable). Run the test in GameMaker first and record actual results before writing. *(raised by Robert; needs hands-on GameMaker testing)*
- ☐ Cross-engine design drift (HUD object, prefabs, helper objects): during the Lesson 10 review Robert noticed that structural pieces introduced in one engine are missing or different in others. Examples: Unity and Unreal got a dedicated HUD (a HUD GameObject / WBP_HUD widget) while Scratch, Godot, GameMaker, GDevelop, TIC-80 and 001 Game Creator draw score and lives directly from the GameManager or from labels; Unity uses Prefabs for enemies while GameMaker's equivalent was never used even though one exists. Before the book, audit every engine for pieces present in some but not all (HUD, prefab/template for enemies/bullets, spawner, restart routine, start_game function), decide per piece whether it is a course-wide concept (then add it to every engine, using each engine's native equivalent) or an engine-specific need (then document the driving reason per Rule 5), and record the result in the Cross-Engine Consistency Reference and Data Dictionary. Add a Standards check: any new structural object introduced for one engine must be introduced for all, or the reason documented. *(raised by Robert at Lesson 10 review; audit needed)*
- ☐ Cross-engine name normalization (generalized): the same object, variable and function must be spelled identically in every engine and every document wherever the engine allows, and the casing must follow the single book convention (PascalCase for assets and objects, per the naming decision above). Known example found at the Lesson 10 review: GameMaker's GameManager object is spelled obj_gamemanager in the shipped handouts but obj_GameManager in Robert's own test project (drift from the late switch to PascalCase). In the book pass, audit and normalize every name (GameManager, game_active, enemies_remaining, lives, score, start_game and the rest) against the Data Dictionary, per engine, then regenerate all Handouts, Keys, Slides, diagrams and the Complete Script Documents from the normalized names. *(raised by Robert at Lesson 10 review; book pass)*

- ☐ Canonical bug-hunt exercise (decision): the win/lose gap (kills reduce enemies_remaining but escapes only reduce lives, so some rounds end with no win and no loss, and R does nothing) is INTENTIONALLY left in Lessons 6 and 8. It is the Lesson 10 Level 3 'Fix a real bug!' task and part of the canonical lesson plan, also for the book. Do not fix it earlier. Plan at least one similar find-and-fix-your-own-design-bug exercise in Semester 2 (Lessons 16-30), where the groundwork is laid in earlier lessons by thinking about two cases separately instead of as one state. *(decision by Robert at Lesson 10 review)*
- ☐ Reconsider static members in Unity's GameManager (Robert): from Lesson 6 on, the Unity GameManager keeps shared state in static fields and static methods (RegisterKill, RegisterEscape, score, lives, gameActive). That is more complicated than the other engines need and forced a Singleton in the first Lesson 10 draft, since a static method cannot use an Inspector reference. Lesson 10 now avoids it with an ordinary instance method ResetGame() plus an Inspector slot for the spawner. For the book, decide whether GameManager should be a normal component with instance fields (other scripts reach it through an Inspector reference, as with the enemy prefab), and retrofit Lessons 6-10. Static members and singletons may then be a good Semester 2 refactoring topic, introduced on purpose. *(raised by Robert at Lesson 10 review; book pass)*
- ☐ Unity GameManager without statics AND without Find...() (book design change, related to the static-members item above): in the book version, avoid FindFirstObjectByType and similar lookups as well as statics. The obstacle is that a runtime-spawned prefab (Enemy, Bullet) cannot hold an Inspector reference to a scene object. Robert believes he solved this in an earlier project: REVIEW HIS OLD PROJECTS for a beginner-level approach before choosing one. Candidates to compare against it: (1) the spawner holds the GameManager reference in its own Inspector slot and hands it to each new enemy right after Instantiate; (2) the GameManager does the spawning itself, so it already holds every reference; (3) events/UnityEvents, the Unity cousin of the Godot signals sidebar (create it, connect it at spawn, handle it on GameManager, invoke it); (4) a shared ScriptableObject holding the game state. Pick the one with the least new vocabulary that fits Lessons 6-10, record the decision, and update Rule 12 and the Data Dictionary so the Unity design matches the other engines. *(raised by Robert at Lesson 10 review; review old projects first)*
- ☐ Design principle, reiterated (cross-engine drift): the game design should mirror itself from engine to engine as much as possible: same objects, same names, same state on the same manager, same flow, same behavior on screen. The one exception is where an engine fundamentally refuses to do it, or where mirroring would need an excessive, unnecessary workaround. In that case the engine's own idiom overrules consistency, and the deviation is recorded with its driving reason (Standards Rules 5 and 12). Apply this test to every item in the drift family: the HUD object, prefabs, the Unity statics and Find...() question, Godot signals versus other patterns, and the Scratch clone and broadcast handling. When one engine needs a heavier pattern than the others (the Lesson 10 Unity Singleton was the example), look first for a simpler design that all engines can share before accepting the deviation. During the book pass, fold this wording into Rule 12 and keep a per-engine list of accepted idiom exceptions in the Cross-Engine Consistency Reference and Data Dictionary. *(raised by Robert at Lesson 10 review; principle for the book pass)*
- ☐ Objects talking to each other: signals, broadcasts and events (add to the Unity design reconsideration above): Godot signals have caused difficulty and blank stares, and the course never gave a proper discussion of what a signal IS before using one. Scratch's Broadcast was glossed over the same way, and Unity events (candidate 3 above) would

raise the same problem. For the book, treat 'one object announces something, another object reacts' as a named concept introduced on purpose, before it is first used (Lesson 6 or earlier, matching the signals-sidebar move already logged), with the same vocabulary and the same picture across engines: Scratch broadcast and receive, Godot signal and connect, Unity event, GameMaker direct calls. Also compare simpler Godot alternatives before settling on signals: the Enemy calls a GameManager function directly through a node path, a Group call (get_tree().call_group), or an Autoload GameManager (Godot's counterpart of the static idea above, so decide it together with the Unity static question). Robert is not sure Godot has a simpler alternative, so verify these against a real project. Pick one pattern per engine, record the reasons, and add the picture to the Rule 9 diagram set. *(raised by Robert at Lesson 10 review; design decision plus a new concept lesson)*

- ☐ Unity static GameManager - revisit in THIS course if needed: the Lesson 6-12 Unity code keeps its static GameManager for now. If a student struggles with it, add a 'Non-static variant' box to the Unity Keys: remove static from the fields and from RegisterKill/RegisterEscape, and in every script that uses GameManager (Enemy.cs, Bullet.cs, the Lesson 12 sound code and so on) add a GameManager field filled in Start() with FindFirstObjectByType<GameManager>(), then change each GameManager.something call to use that field. Watch for the errors 'An object reference is required for a non-static member' and NullReferenceException. *(decision by Robert at Lesson 10 review; do only if needed)*
- ☐ Active-engine scope (set at the Lesson 10 review): the only engines students are currently using are Scratch, Godot, GameMaker (script and Visual) and Unity (script). Until the book pass, new and revised lesson work covers only those. Deferred, not dropped: Unreal, 001 Game Creator, Unity Visual Scripting, and GDevelop and TIC-80 (confirmed paused by Robert). For the book pass, catch each deferred engine up on every Rule 11-15 change, the Lesson 10 restart design, and the Rule 9 diagrams, using the shipped Lesson 10 content (Unreal and 001 Game Creator were already updated in Lesson 10 before this decision) as the starting point. Rendering and re-shipping the deferred engines' Handouts/Keys is paused until then. *(raised by Robert at Lesson 10 review; book pass)*
- ☐ Unity Visual Scripting pass (deferred): no students are currently using the Unity Visual Scripting path, so it was left out of the Lesson 10 pass. For the book, review every Unity Visual section from Lesson 8 on: verify node names against a real Unity install (e.g. the Lesson 10 restart graph and its On Input System Event Button node), apply the Rule 13/14 changes (state on the manager, complete code or graph per level), and build the missing Rule 9 Unity Visual node diagram for Lesson 10 (Lessons 7-9 already have one). *(raised by Robert at Lesson 10 review; book pass)*
- ☐ Lesson 10 compliance pass (done): Handouts and Keys for all eight engines now follow Rules 11, 13, 14 and 15 (shared state on the manager object, explicit loop/counter names, complete highlighted Level 1 code, Key code boxes for Levels 1-3). Rule 8 (slide speaker notes copied from the Teacher's Script, plus a new Activity Diagram slide) and Rule 9 (restart-flow Activity Diagram in all Handouts/Keys/Slides; GameMaker Visual and Unreal Blueprint node diagrams) are also done. Remaining: the GameMaker Visual R-key block is confirmed as 'If Key Pressed' (Letters > R), and the Unity Visual path is deferred (see above); Lessons 11-15 still need the same Rule 13/14 pass. *(Lesson 10 review)*
- ☐ Reflex habits: testing, documentation and refactoring as things students do without thinking of them as separate tasks (Robert's goal, raised at the Lesson 11 review after his own GameMaker project drifted from the Keys): see the separate document Reflex_Habits_Design_Notes.docx (this folder). It holds Robert's problem statement,

Claude's first-pass commentary (yes at beginner level if taught as three plain habits: try it, say it, tidy it, in the last five minutes of lab), a draft candidate Rule 16 (Done Means Tried, Named and Tidy), concrete mechanisms, options for fixing the instructor's own drift, a long spitballing list, risks, and the decisions needed. Nothing in it is adopted yet. *(raised by Robert at the Lesson 11 review; decision pending on candidate Rule 16)*

- ☐ Name the bug: "the Stalemate" (book edition, Robert's decision at the Lesson 11 review): refer to the win/lose gap bug by this name everywhere in the book and course materials from now on. Definition for the manuscript: a round that ends with no win, no loss and no restart, because escapes lower lives but never lower enemies_remaining. The name should appear the first time the bug is reported (Lesson 10's Level 3), be used in Lesson 11's wave discussion, and in any Semester 2 repeat of the bug-hunt exercise. The name also appears in Lesson 10 and Lesson 11 handouts and Teacher's Keys as of Lesson 11 (Lesson 10's own Level 3 text does not name it yet, only the Lesson 11 materials do; decide whether to add the name to Lesson 10's Level 3 in the book). *(decision at the Lesson 11 review; name adopted from Robert)*
- ☐ Lesson 11 compliance pass (done for active engines): Handouts and Keys for Scratch, Godot, GameMaker (script and Visual) and Unity were rewritten to match the Lesson 10 design: wave and total_waves live on the GameManager (Rule 13), the Lesson 10 spawn routine is reused as the 'spawn a wave' routine (spawn_enemies / SpawnEnemies, broadcast [spawn enemies]), complete highlighted Level 1 code (Rule 14), Key code for Levels 1 to 3 (Rule 14), Activity Diagram (new, diagram_activity_wave_flow) and a GameMaker Visual node diagram (Rule 9), game-state check for the new state, speaker notes copied verbatim from the Teacher's Script (Rule 8), Unity demo rewritten without the singleton (Rule 10). The Stalemate is deliberately left in for students to find and fix (Level 3), with a worked wave-aware fix in each Key. Paused engines (Unreal, 001 Game Creator, GDevelop, TIC-80, Unity Visual) were NOT rebuilt: their Lesson 11 files in the lesson folder are the old, stale versions (singleton, spawn_wave naming, no Rule 14 code) and must be rebuilt from the active-engine design during the book pass. *(Lesson 11 review, Rules 8, 9, 10, 11, 12, 13, 14, 15)*
- ☐ Unity static GameManager - concrete cost found in Lesson 11: because RegisterKill() is static, it cannot use the instance Spawner slot, so the wave request is passed through a static flag (nextWaveRequested) that GameManager's Update() acts on. It works and teaches 'static methods cannot reach instance members', but it is exactly the workaround the non-static redesign (see the static-members items above) would remove: with an instance GameManager, RegisterKill() would simply call spawner.SpawnEnemies(). Use this as the motivating example in the book's static-versus-instance discussion. *(found while rebuilding Lesson 11 Unity, Lesson 11 review)*
- ☐ Verify at the keyboard (Lesson 11, not tested by the author): (1) Godot: the next wave is requested with get_parent().call_deferred("spawn_enemies") because a wave clears inside a collision callback (suspected 'flushing queries' restriction on adding physics objects); confirm whether the deferral is actually required or merely harmless, and update the explanation accordingly. (2) GameMaker Visual: the Else block placement after a nested If Variable, and the Draw Value block's field labels, were written from the earlier Lesson 8 to 10 block names and have not been clicked through. (3) Scratch: confirm the last-hit clone never double-spawns a wave (it deletes itself before GameManager's broadcast arrives). *(Lesson 11 review, awaiting Robert's testing)*
- ☐ GameMaker Prefabs: GameMaker has a Prefab concept. Unity introduces Prefabs in the course, so decide where and how GameMaker's equivalent is introduced (likely alongside the lesson where Unity first uses Prefabs), verify what GameMaker Prefabs

actually are and how they differ, and update the cross-engine consistency reference and Data Dictionary accordingly. *(raised by Robert; research needed)*

- ☐ Resolve or explicitly caveat GameMaker's Lesson 13 explosion scale-origin risk before that lesson is written up. *(Task #243, still open)*
- ☐ Finish auditing Unreal Lessons 5, 6, 8, 9, 11, and 14 for the same missing "Split Struct Pin" step found (and fixed) in Lessons 2, 3, and 7. *(Task #254, partially resolved)*
- ☐ When describing Lesson 14 in the manuscript, explicitly call out that its single free-choice Level 3 (breaking the Sound/Visual/Structure template used every lesson since Lesson 2) is intentional scaffolding removal, not an inconsistency. *(found while building Cross_Engine_Consistency_Reference.docx)*
- ☐ Get Robert's decision on 001 Game Creator sprite import steps before that engine's chapter/section can be considered complete. *(Task #209, waiting on Robert)*
- ☐ Before claiming full 8-engine coverage anywhere in the book, finish QA-validating every engine in every mode (visual/script/block), not just the current focus subset (Scratch, Godot, Unity Scripting, GameMaker Scripting, GameMaker Visual, Unreal). *(Task #249)*
- ☐ Revisit lesson pacing across Lessons 3-15 once the full semester has actually run with students - note anywhere the real classroom pace diverged from the written plan before treating the lesson plans as the book's final word. *(Task #119)*
- ☐ Decide whether/how the course's companion digital materials (Handouts, Keys, GDDs) get referenced or reproduced alongside the book - affects how self-contained the book needs to be. *(Task #128, still open)*
- ☐ Check whether IGDA's Game Education SIG has released their updated Curriculum Framework yet, and reconcile the book's framework-alignment claims against it if so - the version available at project start (v3.2 beta, 2008) was already old, and an update was rumored to be in progress. *(see custom project instructions)*
- ☐ Periodically review and consolidate this document's Notes & Commentary section - merge duplicates, resolve contradictions between earlier and later notes, cut anything that turned out not to matter. Do NOT do this reflexively every time a note is added; batch it into occasional dedicated passes. *(standing item - see How To Use This Document)*
- ☐ Write a Lesson 4 Sidebar (or a Rule 6 Refinement and Style section, once that's a course fixture) on "who's in charge" - Bullet and Enemy each detecting their own hit and destroying only themselves, instead of one side reaching into the other. Ties to a live classroom discussion already introduced verbally but not yet formalized in writing. *(raised while checking Unity Lesson 4 sound handling, deliberately deferred rather than retrofitted)*
- ☐ Decide whether the 15 already-completed lessons get a retrospective retrofit pass for Standards Rule 6 (SuperStretch tie-in, Refinement and Style, visual-scripting node-reuse), or whether that content is written fresh for those lessons only when the book manuscript reaches them. See Notes & Commentary for the tradeoffs and a leaning recommendation - not yet decided. *(Standards Rule 6, added this session; scoping only, no retrofit started)*
- ☐ When rebuilding/auditing the Unreal Lessons, verify the Setup instructions actually include creating a no-Pawn Game Mode Blueprint (parent Game Mode Base, Default Pawn Class set to None in its Class Defaults) and setting it as the project's Default Game Mode - and check World Settings > Game Mode Override on each lesson's level too, since an override there takes priority over the project default. Confirmed missing/lost from Robert's own reference project this session: without it, the engine auto-spawns and possesses a stray built-in DefaultPawn at BeginPlay, which caused a hard-to-diagnose

symptom (a black sphere rising out of view, tied to Space being a legacy default "move up" binding on DefaultPawn colliding with Fire also being bound to Space) plus spurious collision-notice firings. *(confirmed via Robert's hands-on Unreal rebuild this session; see also the DefaultPawn note already under Notes & Commentary)*

- ☐ Reminder for Robert, not a course-content item: once Lesson 05 testing starts on his own project, go back and add Lesson 4's Color Vector Parameter tint / Lesson 5's striped-row idea into his own per-engine test projects - he skipped it during Lesson 4 testing as optional/untested, but confirmed (by reading the actual Lesson 4/5 Unreal content) that it's genuine forward-designed setup, not filler. *(personal test-project follow-up, not a document edit)*
- ☐ Reminder for Robert, not a course-content item: re-check his Scratch project's two enemy boundary values (the escape check at y position < -180 instead of the Key's -170, and the enemy spawn y of 130 instead of the Key's 150) by actually playtesting at full screen rather than only in the editor's small inset play window, since both were flagged as "observed placement" that may read differently at a larger render size. Worth noting for when he does: Scratch's own stage coordinate system (x: -240 to 240, y: -180 to 180) is fixed regardless of window size - only the rendered pixel scale changes, not the logical boundaries - so if -180 does cause a visible pop, it is equally true in both the small window and full screen; what's likely to change with screen size is how NOTICEABLE a small pop is, not whether the numbers themselves are correct. Worth keeping both possibilities open when he retests. *(personal test-project follow-up, not a document edit; Scratch coordinate-system clarification included for when Robert revisits it)*
- ☐ Retrofit past lessons' visual-scripting alternate-path Handouts and Teacher's Keys against Standards Rule 7 (name every block, category, and field - not just the shape of the logic). GameMaker's Lesson 5 GML Visual altPath was the first case caught and fixed, while testing his own reference project surfaced two real gaps: the setup step didn't distinguish deleting a room instance from deleting the object asset, and the altPath's note wrongly implied the Level 3 color-switch idea needed no changes from the code path when it actually needed its own block sequence. Audit every other lesson's GML Visual, Unity Visual Scripting, GDevelop, 001 Game Creator, and Unreal Blueprint alt-path content the same way - unlike Rule 6, Robert has directed this one to be retrofitted, not deferred to book-writing time. *(Standards Rule 7, added this session per Robert's explicit direction; Lesson 5 GameMaker, Lesson 8 (GameMaker/Unity altPath field/node explicitness), and Lesson 9 (Unity Visual Scripting altPath node-name explicitness) done, Lessons 2-4, 6-7, and 10-15 not yet audited)*
- ☐ Retrofit Lessons 1-4, 7, and 10-15's existing Slide Decks (.pptx) with Teacher's Script content in PowerPoint speaker notes, per Standards Rule 8. Lesson 5's deck already has this (each content slide's addNotes() carries that slide's matching Teacher's Script section verbatim, including bracketed stage directions) and is the reference example to follow; Lessons 6, 8, and 9 now also have full coverage (each lesson's six new notes calls follow the same slides-5/7/9/11/13/14 pattern Lesson 6 set, skipping title/overview/divider slides). Robert's classroom now runs Presenter View across two displays, so the notes pane doubles as his live script during lecture - this is a genuine workflow need, not just a nice-to-have, but Robert explicitly deferred the retrofit rather than asking for it done immediately. *(Standards Rule 8, added this session per Robert's explicit direction; Lessons 5, 6, 8, and 9 done, Lessons 1-4, 7, and 10-15 not yet started)*
- ☐ Retrofit Lessons 1-4, 6, and 10-15 against Standards Rule 9 (diagrams): add the generic Activity Diagram to every already-shipped Handout, Key, and Slide Deck; add engine-specific node diagrams to GameMaker/Unity/Unreal's Handouts (Level 1, unconditional) and Keys (Level 1 always, plus a combined Level 3 sequence diagram

where the visual path is complex enough to warrant one); and add diagrams to existing Sidebars where they'd genuinely clarify the text, per the same rule's Sidebars/Slides guidance. *(Standards Rule 9, added this session per Robert's explicit direction; Lessons 5, 7, 8, and 9 now have this treatment, Lessons 1-4, 6, and 10-15 not yet retrofitted)*

- ☐ Apply Standards Rule 9's diagram standard to Lessons 16-30 as each is authored, the same way it's now required for Lesson 5 onward - don't let the not-yet-built future lessons slip back to the pre-Rule-9 no-diagram default. *(Standards Rule 9, forward-application reminder)*
- ☐ Expand the Lessons 5-14 lecture-demo rotation (Standards Rule 10) from its current 5-choice pool (Scratch, GameMaker script, GameMaker visual, Unity C#, Godot) to evenly cover all 8 engines and all 3 coding styles, once students are actually using the currently-unused engines (GDevelop, TIC-80, 001 Game Creator, Unity Visual Scripting, Unreal Blueprints - none have active students right now). The original 30-Lesson Master Schedule already models what full-diversity balance looks like and can be adapted rather than redesigned from scratch. Until then, the restricted 5-choice rotation is intentional, not an oversight. *(Standards Rule 10, added this session per Robert's explicit direction to narrow the pool for now)*
- ☐ Retrofit past lessons' code and visual-scripting content against Standards Rule 11 (meaningful variable names over generic i/j/k counters), across all engines and coding styles. Lesson 5's audit is now complete across all 8 engines: GameMaker (Code and Visual paths, i -> enemy_count, inst -> new_enemy), Godot (Code path, i -> enemy_count, e -> new_enemy, plus a ternary -> if/else cleanup), Unity (Code path, i -> enemyIndex, e -> newEnemy, ternary -> if/else), and TIC-80 (Lua, i -> enemy_count in the spawn loop only - the separate bullets-iteration and enemies-array-iteration loops use i/j for a different concept and were deliberately left untouched) all needed and got a rename. Unreal's C++ alt-path (i -> EnemyCount) also needed one; its Blueprint path needs none (the bare "Index" it uses is the ForLoop node's own fixed pin name, not a variable this course chose - same as Unity Visual Scripting's Index pin). GDevelop and 001 Game Creator were both audited and found to need no change, for two different reasons: GDevelop's self-declared EnemyIndex scene variable was never generic to begin with; 001 Game Creator has no loop or variable at all for this concept - Level 1 is pure manual copy/paste placement in the Scene editor, and Level 3's row variety is manual per-instance Tint editing, so there's nothing to rename. Lesson 5 is done; every other lesson (1-4, 6-15) has not yet been audited for the same generic-counter pattern. Robert has adopted a standing workflow for this going forward: rather than a dedicated retrofit sweep, run a quick Rule 11 naming pass on an engine's Lesson 5+ content the next time he personally starts testing that lesson/engine combination, opportunistically, as he works through the course. *(Standards Rule 11, added this session per Robert's explicit direction; Lesson 5 fully audited across all 8 engines (GameMaker/Godot/Unity/Unreal C++ renamed, Unreal Blueprint/GDevelop/001GC confirmed no change needed, TIC-80 renamed), opportunistic per-lesson-as-tested workflow adopted for Lessons 1-4/6-15 in place of a dedicated retrofit sweep)*
- ☐ Pick a specific Semester 2 lesson to formally introduce the ternary operator, once Semester 2's actual lesson content is being built - Robert's own suggestion, prompted by pulling the ternary out of Lesson 5's GameMaker code for being too advanced this early. Not yet scheduled to a lesson number; needs to land somewhere a student is already comfortable with plain if/else and is extending an existing codebase, not meeting conditionals for the first time. *(raised alongside the Lesson 5 ternary removal; see the matching Notes & Commentary entry)*
- ☐ Rename Scratch's Lesson 5 loop variable from enemy_number to enemy_count, to match GameMaker and Godot. Robert's decision: enemy_count wins the naming

question outright (it already had two engines using it before this was even raised as a question), so Scratch - the outlier, and the only engine using enemy_number - gets reconciled to match rather than the other two being changed. Not yet done as of this entry. *(Robert's explicit decision, given while Unity's Rule 11 retrofit was underway; supersedes the earlier not-yet-decided framing of this same question)*

- ☐ Run a dedicated, coordinated pass reconciling BOTH variable/function naming AND code/graph structure across all 8 engines and all 3 coding styles, per new Standards Rule 12 - broader in scope than the existing opportunistic, per-lesson-as-tested Rule 11 naming workflow (above), which only catches naming, one engine at a time, as Robert happens to test it. This pass should also resolve every deliberately-deferred word-choice and structural conflict surfaced along the way rather than fixing any one of them in isolation. Known cases so far, all in Lesson 5's spawn-loop concept alone: Unity's enemyCount (existing total-to-spawn field) vs. enemyIndex (the Rule-11-compliant but non-matching loop counter name used to avoid shadowing it); Scratch's enemy_number vs. the enemy_count used everywhere else (Robert's decision: enemy_count wins, tracked as its own ToDo above); GDevelop's EnemyIndex, meaningful but a different word choice ("index") than the "count" used by GameMaker/Godot/Scratch-once-renamed; and Unreal Blueprint's spawned-actor reference, which is never promoted to a named variable at all (wired directly from Spawn Actor from Class's Return Value), unlike every other engine's named new_enemy/newEnemy - a structural gap, not just a naming one. Also now known: which object owns bullet/enemy collision detection and response is opposite between GameMaker and Godot - GameMaker's obj_bullet has the Collision event and destroys the enemy (and itself) from the bullet's side, while Godot's enemy.tscn owns an Area2D Hitbox (area_entered signal) and destroys the bullet (and itself) from the enemy's side. Neither is wrong on its own, but nobody ever decided which object should own this across engines - it just landed differently as each lesson was authored. Robert's own refactoring preference (each object cleans up after itself / is self-reporting, surfaced while reviewing his GameMaker and Godot projects) points toward a third option worth considering in the eventual pass - the thing that dies announces its own death - rather than simply picking one engine's existing side to standardize on. Robert's explicit decision when this was raised: track it only, do not unwind it for current students now - that reconciliation is real but belongs entirely to this eventual coordinated pass. Also now known, found while Robert reconciled his own Godot project: spawn_enemies() has the same NAME in GameMaker and Godot but a different OWNER - GameMaker's is scr_spawn_enemies, a free-standing global Script resource belonging to no object, callable directly by anything (the bullet, start_game(), wave_cleared()); Godot's is a method on main.gd, owned specifically by the Main scene node, with GameManager having to reach it via get_parent().spawn_enemies(). This is tangled up with a second asymmetry that cuts both ways: Godot has no parallel to GameMaker's obj_spawner, but GameMaker equally has no parallel to Godot's main.gd - GameMaker has no scripted scene-root object at all (the Room is unscripted; obj_gamemanager is just a peer object in it), so main.gd isn't really standing in for obj_spawner, it's filling a role (scene-root ownership of the spawn routine) that doesn't exist as a GameMaker concept at all. Note this partly resolves on the GameMaker side once the planned obj_spawner retirement (see the six-candidate-refactorings ToDo item) happens - at that point GameMaker's spawn-related logic becomes entirely global-Script-resource-based with nothing object-owned, while Godot's would still be split across two node scripts - a real design-philosophy fork (centralize all shared logic on one manager, vs. let each engine's own scene-ownership idiom decide where code lives) worth its own discussion in the eventual Rule 12 write-up, separate from simple naming mismatches. Robert has also asked, once this reconciliation pass happens, to

extend the same alignment effort to Scratch's objects/events/broadcasts - matching Scratch's existing GameManager sprite and broadcast-message names to the equivalent function/event names used in the other 7 engines as closely as Scratch's own block-based model allows, acknowledging this may land more as explicit cross-references in the Data Dictionary (broadcast message X = function Y in engines A/B/C) than as literal renaming, since Scratch's own mechanisms won't map one-to-one. Updated again to record that Robert's own GameMaker/Godot ownership-reassignment drift (see the dedicated ToDo item above) is the same underlying pattern as this collision-ownership asymmetry for those two engines - Robert's own projects actively explored both directions (bullet-owns vs. enemy-owns) before this Rule 12 question was ever raised formally. Updated again to add a FOURTH engine data point, found while Robert reconciled his own Unity project: his Bullet.cs's Start() plays its own fire/shoot sound via a private AudioSource field - a deliberate ownership choice, confirmed explicitly by Robert and kept as-is (not reconciled to canonical's Player-owns-the-shot-sound placement) specifically so it can be documented later for the "Doing It the Doers of Stuff Way" chapter - and his Enemy.cs had taken over OnTriggerEnter2D collision detection/response (RegisterKill(), hit sound, destroying both objects) that canonical keeps exclusively on Bullet.cs, the same bullet-vs-enemy ownership swap already tracked for GameMaker and Godot above. Unlike the fire-sound placement, Robert asked for the collision-ownership split to be reconciled back to canonical for the current course version, which was done (collision detection, the hit AudioClip, and RegisterKill() moved back onto Bullet.cs; Enemy.cs now only moves, checks the off-screen threshold, and calls RegisterEscape()). CORRECTION, same session: Scratch is NOT an example of Robert's drift the way GameMaker and Godot are - Scratch's own shipped Key deliberately switches from bullet-owns (Lessons 3-4) to enemy-owns (Lesson 5 onward) ON PURPOSE, for a genuine Scratch-specific technical reason stated directly in the Lesson 5 Key: Scratch clones have no unique ID, so only the Enemy clone being touched can identify and delete itself - the Bullet side has no way to know which specific Enemy clone it hit. This makes Scratch a THIRD real data point for this list (enemy-owns, like Godot, but for a completely different and well-documented reason), and it also means Scratch's own Lesson 3-4 vs. Lesson 5+ materials are themselves internally inconsistent about which object the Handouts/Keys show owning collision - worth flagging as its own small audit item (do Lessons 3-4's materials still read as if that design persists forward, or do they already set reader expectations correctly for the Lesson 5 change) during the eventual Rule 12 pass. Robert's stated reasoning: the goal throughout is enabling students on different engines to help each other, which is worth pursuing mindfully even where a perfect one-to-one mapping isn't possible. This pass should also settle the open PascalCase-vs-camelCase-vs-snake_case question once and for all (see the separate ToDo item on that question) so the coordinated naming pass has one settled convention to apply rather than re-litigating case style engine by engine. Updated again to record a new, confirmed case found during Lesson 14's compliance pass: TIC-80 genuinely and correctly calls its spawn routine spawn_wave() - NOT spawn_enemies() like Godot, GameMaker, and Scratch, and NOT SpawnEnemies() like Unity. This was discovered specifically because fixing a spawn_wave()-vs-spawn_enemies() drift bug that affected four OTHER engines required definitively confirming TIC-80's own correct name before leaving it alone, which surfaced a real, previously undocumented two-way split (spawn_wave vs. spawn_enemies/SpawnEnemies) worth folding into the eventual coordinated pass alongside the existing cases. *(Standards Rule 12, added this session per Robert's explicit direction; a fundamental precept (common ground across engines so students can help each other regardless of engine/style), not yet started; growing list of known reconciliation cases tracked here as they're found)*

- ☐ Decide the course's cross-engine variable/function naming CASE convention (PascalCase vs. camelCase vs. snake_case vs. some other scheme) as part of the Standards Rule 12 reconciliation pass (above) - a distinct question from which underlying WORD an engine uses (e.g., enemy_count vs. enemy_number), which is about the letter-casing/formatting style layered on top of whichever word is chosen. Robert's stated inclination: favor cross-engine consistency in the actual case/format used, even in engines where a different case convention is the idiomatic local norm (e.g., Unreal/C++ typically PascalCase, C-family languages typically camelCase, GML/GDScript typically snake_case) - and where the chosen course-wide convention deviates from an engine's own idiom, explain that deviation explicitly in a Sidebar rather than silently picking the engine-idiomatic case and letting the inconsistency stand unexplained. Not yet decided which specific convention wins; this entry only records the framing and Robert's leaning, not a final choice. *(raised alongside the Data Dictionary discussion; Robert's explicit inclination given, final convention choice deferred to the Rule 12 pass)*
- ☐ TIC-80's bare loop variables j and e (the enemy-array iteration pattern, e.g. "for j=1,#enemies do" / "local e=enemies[j]") were renamed to enemy_index/enemy for Lessons 7 and 8 as part of each lesson's own compliance pass. The identical unrenamed pattern was confirmed, by direct grep, still present in Lessons 5 and 6's build scripts. Robert's lesson-scoped authorizations for those passes didn't extend to Lessons 5-6, so this was deliberately left as a follow-up rather than done opportunistically alongside them. Fold into either a dedicated Rule 11 TIC-80 sweep or the broader Rule 12 reconciliation pass (above), whichever comes first. *(found during Lesson 7's Rule 13/diagram compliance pass, then confirmed already fixed for Lesson 8 during that lesson's own pass; Lessons 5-6 still open)*
- ☐ Decide, for the book, whether Unity's early lessons should introduce [SerializeField] private fields instead of (or alongside) plain public fields for Inspector-exposed values. Raised while Robert reconciled his own Unity project (EnemySpawner.cs uses [SerializeField] private enemyPrefab/enemyCount where canonical uses public). Robert's explicit framing: he changed it in his own project because he knows the distinction, but deliberately did NOT ask for the canonical course materials to be changed, because he isn't convinced the encapsulation conversation is worth it for a beginner class - it may belong in the "Doing It the Doers of Stuff Way" refactoring chapter instead, or nowhere at this course's level. His current leaning: keep canonical on plain public fields for simplicity, but he wants this decided intentionally, as its own discussion, not defaulted into either way. Not yet decided. *(raised while reconciling Robert's EnemySpawner.cs; Robert's own project differs from canonical on this point by his own deliberate, informed choice - not drift to fix)*
- ☐ Reminder for Robert, not a course-content item: re-drag the Inspector field references (spriteDefault/Left/Right/Up/Down, sr, walkingBeep, xBoundary, yBoundary) onto the reconciled PlayerMovement.cs and playtest - the restructured single-moving-check version (fixing the diagonal-movement blink-rate bug and the missing walkingBeep.Stop()) has not yet been verified at the keyboard. *(personal test-project follow-up, not a document edit; last file in the Unity reconciliation sequence)*
- ☐ Fix the shipped Lesson 4 Unity Handout/Key for the book: AudioSource.PlayClipAtPoint(hitSound, transform.position) should be AudioSource.PlayClipAtPoint(hitSound, Camera.main.transform.position). This is a genuine, confirmed bug (PlayClipAtPoint() always creates a 3D-blend temporary audio source with real distance rolloff, so a sound played at the enemy's own position is quietly, almost inaudibly wrong, unlike every other PlayClipAtPoint call in the course, all of which already use Camera.main.transform.position) - found while Robert debugged his own quiet hit sound. Robert's explicit decision: do NOT fix the live Lesson 4 materials

now, mark for book correction only. Lesson 13 forward was checked and fixed separately (see the matching Notes & Commentary entry) since it's live, current content. *(confirmed bug, found via Robert's own project; Lesson 4 itself deliberately left unfixed per Robert's instruction, Lesson 13 already fixed)*

- ☐ Write a Unity troubleshooting-sidebar entry on AudioSource.PlayClipAtPoint()'s quiet-sound trap: it always spawns its temporary audio source with 3D spatial blend and real distance rolloff, so a sound played anywhere other than Camera.main.transform.position (or very close to the camera) can end up nearly inaudible - and the optional volumeScale argument can't fix it, since it's a 0-1 percentage of the clip's own volume, not a boost, and can never compensate for rolloff. The practical fix used throughout this course is simply to always call PlayClipAtPoint at Camera.main.transform.position, the same pattern GameManager's round-ending sounds already used from the start. Good candidate for the same category of "common pitfalls" sidebar content already tracked for the color-multiply-by-white bug. *(raised while fixing Robert's quiet Bullet.cs hit sound and the matching Lesson 4/13 course bug)*
- ☐ Fix a genuine continuity bug in the shipped Lesson 13 Unity materials: the content describes hit-detection as having always lived on "the Enemy prefab's script," but canonical Lessons 4-12 confirm it has always been on Bullet.cs. Missed during Lesson 13's own compliance pass; found while reconciling Robert's personal Unity project back to canonical (collision ownership there had also drifted onto Enemy, unrelated to this specific wording bug). Low priority - doesn't block any current work. *(found while reconciling Robert's Unity Bullet.cs/Enemy.cs against canonical; not yet fixed)*
- ☐ Standards Rule 13 (dedicated manager object for shared/global state) has only been retrofitted into Lessons 6, 7, 8, and 9 so far (Scratch, Godot, GameMaker, and Unreal gained a GameManager-equivalent object; Unity already had one, which is what prompted the rule; Lesson 8 additionally retrofitted the lives/has_lost lose-condition state onto the same object, and Lesson 9 retrofitted score the same way across GameMaker, Scratch, Godot, and Unreal). Lessons 10-15 haven't been audited yet for the same pattern - any lesson whose Handouts/Keys have a spawner, player, or other object directly owning a shared counter (a win/lose flag, a score, a lives count) rather than routing it through a dedicated manager object is a Rule 13 candidate. Audit opportunistically, the same adopted workflow as Rule 11's naming retrofit, rather than as a dedicated sweep, unless Robert says otherwise. *(Standards Rule 13, added this session; Lessons 6-9 done, Lessons 10-15 not yet audited)*
- ☐ Godot-specific (no real parallel in the other 7 engines, per Robert): decide, for the book, which way of connecting a signal is better/more idiomatic/more consistent/easier to track and debug - writing the connection explicitly in code (e.g. $Hitbox.area_entered.connect(_on_hitbox_area_entered) inside _ready(), the way the shipped Keys show it) versus wiring it point-and-click through the Editor's Signals tab (which Robert's own project uses instead - the connection then lives in the scene file's metadata, not as a visible line in the script at all). Raised while Robert reconciled his own Godot project's enemy.gd against the Lesson 6 Key and noticed his version had no .connect() line because he'd wired it through the Signals tab instead. Explicitly deferred - track only for now, do not change the live course's current approach (code-based .connect(), as already shipped) while reconciling Robert's own project. *(raised by Robert while reconciling his Godot project; Godot-only, no cross-engine parallel; deferred to the book pass)*
- ☐ For the book manuscript, move the GameManager mini-lecture ("Before We Start: Meet the GameManager" / "Why a Dedicated Manager Object?") from where it's live-taught to where it actually belongs conceptually: Lecture 06, not Lecture 07. The two slides live in Lesson 7's deck right now purely for production-timeline reasons - the

retrofit was designed and caught after Lesson 6 already shipped, so it had to land as a lab-time catch-up notice in the next lesson taught, not the lesson the object was actually introduced in. The book isn't bound by that same live-classroom constraint, so this content should be written into Lecture 06's chapter instead, presented as part of introducing the win condition and shared state in the first place, with Lesson 7's chapter only needing a brief backward reference rather than the full explanation. Keep the two slides in Lesson 7's live deck as-is; this is a book-only reorganization, not a live-course change. *(raised alongside adding the GameManager mini-lecture slides to Lesson 7's deck; book-only reorganization, live deck unaffected)*

- ☐ New companion book/class concept, spitballed, not scoped or scheduled: "Debugging 101" - a course/book built around a pre-loaded code repository seeded with a variety of deliberately planted bugs, modeled directly on real categories this very project has hit: flat-out broken code that doesn't compile/run, failed edge cases, a new feature that breaks an existing one, a mismatch between the design doc and the actual code, a broken refactoring, and a misinterpreted feature (built to spec as written, but not to what was actually meant). Natural companion idea: "Testing 101" - the inverse situation, an existing codebase plus a design doc but NO test cases, where the student's job is to build the test suite from scratch. A possible third, combined course: "Testing/Debugging 201", merging both - use the act of writing test cases as the mechanism that both finds the bugs AND proves each fix actually worked (red test before the fix, green test after), rather than teaching testing and debugging as separate skills. All three are genuinely downstream of THIS project specifically: the bug categories map directly onto real bugs found while building this course (see the Notes & Commentary entries on the spawn-loop self-collision bug, the GameMaker Visual targeting saga, the dropped Level 3 sounds, the WaveAnnouncementText/WaveText naming mismatch, and the Stalemate itself), which is exactly the kind of real-world-grounded material this idea would need. Not scoped, not scheduled, no engine/language/audience decided yet - purely a note-it-before-it's-lost entry; see the matching Notes & Commentary entry for the fuller framing. *(Robert's idea, spitballed; companion concept to the main Video Game Creation course, not yet scoped)*
- ☐ GameMaker GML Visual audit: confirm whether any existing Visual-path content assumes an "Else If" node/block exists. It appears GML Visual has no such block - a three-or-more-way branch (if / else-if / else-if / else) has to be built by nesting a second If Variable (or equivalent) block inside the first one's Else slot, one level deeper each time, rather than chaining same-level blocks the way code's else if does. Any Handout, Key, or diagram instructing a student to "add an Else If block" in GML Visual is describing a block that may not exist and needs rewriting as nested If/Else instead; any node diagram showing a flat else-if chain for GameMaker Visual needs the same correction, drawn as actual nesting. Not yet audited against the shipped lessons - verify at the keyboard which lessons' GML Visual content has a 3-plus-way branch (Lessons 8-9's lives-vs-enemies-remaining check and Lesson 11's wave-cleared check are the most likely candidates) and confirm whether they already nest correctly or need fixing. *(raised by Robert; verify at the keyboard before concluding which lessons are affected)*
- ☐ Extend Standards Rule 12 (cross-engine consistency) to cover pop-up/overlay text presentation, not just variable and function naming: wave announcements, win/lose banners, and other HUD or momentary-message text should look as similar as reasonably possible from engine to engine - specifically including things like centering on screen, vertical placement, and text size, which the course has so far let drift silently (Robert's own words: "I've silently deviated on a few things like text placement, size, and such"). Proposed shape of the fix: define one simple, explicit set of reasonable defaults (e.g., centered horizontally, a consistent vertical band for a given message role, a

consistent relative size for "HUD counter" text vs. "big announcement" text) and apply it as the baseline in every engine's Handout/Key, while still leaving room for a student to deviate on these small things if they want to. Robert's explicit design goal for how this should read in the lesson materials: it should take very little callout in the lesson plans themselves - the consistency should mostly just BE there, quietly, rather than being taught as its own topic - while still being fairly obvious to a student who looks at the actual steps what the convention is and why those particular values were chosen. A secondary benefit Robert flagged on purpose: nailing down one specific way to do a small, otherwise-you'd-have-to-look-it-up task (centering text on screen in each engine, for instance) functions as a tiny, free lesson in that engine's own text/UI API that the course doesn't have to spend any lecture time on, but that saves a student from going and searching for it themselves. Needs: an actual audit of every existing pop-up/HUD text element across the shipped lessons to find where defaults have already silently diverged, a decided set of default values per message role, and a lightweight way to state the convention in each engine's Setup/instructions without it reading as a big teaching moment. *(raised by Robert; extends Rule 12's existing scope from naming/structure to visual presentation of HUD/overlay text; not yet audited or decided)*

- ☐ Book retrofit (Robert's decision, raised while reviewing Lesson 12): remove ALL sound assets and sound programming currently scattered as optional Level 3 ideas across the earlier lessons (the movement blip in Lesson 2, the shoot sound in Lesson 3, the hit/explosion sound in Lesson 4, the victory sound in Lesson 6, the life-lost/game-over sounds in Lesson 8), and reserve sound entirely for a dedicated Lesson 12 that teaches it properly as its own concept, not as a scattered series of optional add-ons. Robert may still choose to keep some of these as stretch/Level 3 goals in individual earlier lessons, but the explicit design intent either way is to make a conscious effort to "pull it all together" in Lesson 12 rather than let sound arrive piecemeal. This is a book-only retrofit - the live course's Lessons 2-11 already shipped with the scattered-optional-sound design and are not being changed retroactively; Lesson 12 itself was already corrected this session to assume every one of those sounds might be either freshly built or already present (see the Lesson 12 compliance pass note above), so the live course's actual Lesson 12 content does not need to change for this decision - only the earlier lessons' content would, in the book version. See the matching Notes & Commentary entry for Robert's full reasoning and its connection to the Level 1/2/3 framework question also raised in this same message. *(Robert's decision, raised reviewing Lesson 12; book retrofit only, live course Lessons 2-11 unaffected)*
- ☐ Clarify and document the intended PURPOSE distinction between Level 1 and Level 2 across the whole course (Robert's framing, raised alongside the sound-consolidation decision above): Level 1 (Minimal/MVP) exists specifically so a classroom-paced student has a safety net - something they can reach quickly enough to not fall so far behind the live class cadence that they can no longer reasonably absorb the next lesson. Level 2 (Complete) is the actual PREFERRED accomplishment level, especially for self-paced/individual study, because it represents a more realistic, complete feature set and deliberately mixes "spoon-fed" instruction with "think on your own" challenge in a way Level 1 alone does not. This reframes Level 1 as a classroom-pacing safety valve rather than the course's real target, with Level 2 being the actual goal the course is designed around - worth stating explicitly in the book's framework chapter (see the existing "Level 1/2/3 framework as the book's spine" Notes & Commentary entry) rather than leaving the Level 1 vs. Level 2 relationship implicit. A direct consequence for the sound-consolidation decision above: Robert is fairly sure Level 2 should NOT include sound effects once sound is consolidated into Lesson 12, since Level 2 is the

realistic/preferred target students should reach across the earlier lessons, and sound doesn't belong there until its own dedicated lesson - not yet finalized, but stated as a strong leaning. *(Robert's framing, raised alongside the sound-consolidation decision; affects how Level 1/2/3 goals are worded across the book, not yet drafted)*

- ☐ Six candidate refactorings for the "Doing It the Doers of Stuff Way" final chapter (Robert, spotted while reconciling his own GameMaker Script project back into compliance with the shipped Lesson 12 code): (1) enemies_remaining should be incremented directly inside each enemy's own Create event rather than set via a fixed count inside the spawn loop that creates them - lets each enemy be self-reporting rather than relying on the spawner to know the total. (2) The number of enemies to spawn (currently a "mystery number" - 5 - hardcoded inside the spawn loop) should instead be a named variable (something like enemies_per_wave), so the value's meaning is visible at the point it's used, not just inferable from context. (3) enemies_remaining checks/decrements are currently scattered across multiple locations (obj_bullet's Collision event, obj_enemy's Step event, both now also routed through wave_cleared()) - Robert thinks most or all of this logic should consolidate onto a single, dedicated location: obj_enemy's own Destroy event, so that however an enemy goes away (hit or escape), the SAME one place is responsible for updating the count, rather than each removal path updating it independently. (4) The player-lives check (currently __dnd_lives, read/written from obj_enemy's Step event on an escape) arguably doesn't belong on obj_enemy at all - Robert leans toward it living on either obj_player or obj_gamemanager instead, since lives is fundamentally about the player/game state, not something the enemy itself should be reaching in to modify. (5) obj_player's Step event has accumulated enough logic over the semester (movement, firing, and whatever else has landed there) that moving movement specifically into its own separate script would reduce clutter and give Step a single, easier-to-scan responsibility. (6) obj_spawner's sole remaining job as of Lesson 10 is a single-line Create event (start_game();) - eventually that one call could move onto obj_gamemanager's own Create event instead, retiring obj_spawner as a distinct object entirely, since it no longer does anything a GameManager couldn't just as easily do for itself. Robert specifically noted that obj_spawner's own HISTORY across the course - starting as the object that owns the entire spawn loop (Lesson 5), shrinking to a single delegated function call once start_game() existed as its own Script resource (Lesson 10), and potentially being retired altogether as a final step - is itself a good, concrete, already-real example of incremental refactoring for the final chapter to walk through, not just the end-state change. None of these six are being implemented now - explicitly captured as raw material for the eventual dedicated refactoring-walkthrough chapter, where the WHY behind each one (self-reporting objects, named vs. magic numbers, single point of responsibility for a shared counter, state living on the object it actually belongs to, extracting a script to reduce event clutter, and retiring an object once its responsibilities have all been absorbed elsewhere) can get the full discussion Robert wants that chapter to have. Confirmed cross-engine while reconciling his Scratch project: Robert had already independently built candidate (1) (self-reporting enemy count) there too - each Enemy clone broadcast its own creation to GameManager, which incremented enemies_remaining per-clone rather than a flat set-to-5 - and said he "probably did this in every engine." This is being reconciled back to the canonical flat-set design in the live course materials (not kept), per Robert's explicit decision, but confirms the idea wasn't a one-off thought - he built toward it independently and repeatedly before it was ever written down here, which is itself worth mentioning when the final chapter discusses candidate (1). *(Robert's own refactoring observations, raised while reconciling his GameMaker project; explicitly for the final refactoring chapter, not implemented now; item 6 doubles as a real example of*

incremental (not one-shot) refactoring; candidate (1) independently confirmed built (then reconciled away) in Scratch too)

- ☐ GameMaker Lesson 12 sound calls given at the Level 1 architecture (inline win/wave-advance if/else in obj_bullet, direct spawn_enemies() call) may need reconciling against a project that's already past Level 1 - specifically, one that has the Lesson 10/11 Level 3 "Stalemate" fix in place (a shared wave_cleared() function, called from both obj_bullet's hit path and obj_enemy's escape path, so an escape can also clear a wave). If wave_cleared() already exists in a given project, Lesson 12's sfx_wave call belongs ONCE, inside wave_cleared()'s else branch, not duplicated/left inline in obj_bullet's Collision event. Raised while Robert was reconciling his own GameMaker Script project (which has deviated furthest from the shipped course materials of any engine) against the corrected Lesson 12 Key. Explicit decision: do NOT retrofit this now, in either the live course's shipped Lesson 12 materials or Robert's own test project. Leave it be, and handle it one of two ways later - either as an explicit refactor if a later lesson's own progression genuinely requires the wave_cleared() consolidation to exist by that point, or fold it into the planned "Doing It the Doers of Stuff Way" final refactoring chapter as a worked example (a shared-function extraction prompted by a second call site appearing, which is a very natural refactoring story to walk through). *(raised while Robert reconciled his own drifted GameMaker project against the Lesson 12 Key; explicitly not retrofitted now)*
- ☐ NEW Lesson 15 deliverable, fully scoped and decided (Robert), but explicitly DEFERRED - do not start building yet: a "Complete Codebase" document per engine, handed to students at Lesson 15 so they can check their own work over the semester break and come back ready for Semester 2. Scope, as decided: (1) Engines - the active 5 (Scratch, Godot, Unity C#, GameMaker Code, and GameMaker Visual), matching the existing active-engine-scope decision; the 4 paused engines (GDevelop, TIC-80, 001 Game Creator, Unreal) are explicitly out of scope for this deliverable. (2) Depth - NOT just the reliable Level 1/2 baseline; Level 3 "complete code" should be included wherever it exists, which means first BACKFILLING the known gap where several lessons' Keys only describe Level 3 in talking-point prose rather than full code (see the Rule 14 retrofit ToDo item), rather than skipping those lessons' Level 3 content. Robert's explicit reasoning for choosing the harder option: doing that backfill work now feeds forward into the book version too, so it isn't wasted effort scoped only to this one deliverable. (3) GameMaker Visual - represented as full node-flow diagrams per object, not prose description, extending the existing wireframe-diagram pilot (Lessons 5/7/8/9/11) to cover every GameMaker Visual object/script in the finished project completely, end to end. CRITICALLY: Robert explicitly said NOT to start this now - wait until the remaining lessons (through at least Lesson 15) have been reviewed first, because building this deliverable before that review is likely wasted effort (the reviewed lessons may still change, which would mean redoing the Complete Codebase work). Treat this as a planned, fully-scoped future task, not a current one - revisit once the remaining-lessons review is done. *(Robert's decision, fully scoped; explicitly deferred until the remaining lessons are reviewed - do not start early)*
- ☐ GameMaker naming inconsistency found and fixed: the drag-and-drop sound-playing block is called "Play Audio" (confirmed by Robert directly against his own installed GameMaker version). Several shipped Lessons (8, 10, 11, and the newly-added Lesson 12 content) had instead called it "Play Sound" or "Audio Play Sound" - genuine naming drift, not a deliberate choice. Fixed across build_lesson08/10/11/12_handouts_keys_checklist.js, build_lesson12_plan_teacher_script.js, and build_lesson12_slides_deck.js; rebuilt, PDF-verified (grep confirmed "Play Audio" present, no stale text remaining), and shipped to the Lesson 8/10/11/12 folders. Lesson 13's similar-looking "Play Sound"/"Play Sound

2D" references were checked and confirmed to belong to 001 Game Creator and Unreal Blueprint respectively - different engines' real terminology, correctly left untouched. *(Robert confirmed the correct block name at the keyboard; fix completed and shipped this session)*

- ☐ Book-only final chapter, decided (Robert): "Doing It the Doers of Stuff Way" - a dedicated, one-time refactoring walkthrough, taking the capstone's finished Semester 1 project and refactoring it end to end with full discussion of why and how, introducing Robert's own particular coding style and opinions explicitly as HIS style, not as a new ongoing requirement. This explicitly REPLACES an earlier, now-abandoned idea of a standing "Level 4" (refactoring + personal style, applied lesson by lesson alongside Levels 1-3 the same way Level 3/stretch goals and Sidebars already are). Robert's stated reason for rejecting the ongoing-Level-4 approach: it would force a choice between two bad options every single lesson going forward - either make refactoring-to-Robert's-style a mandatory part of every lesson (which he does not want), or maintain two parallel code paths per lesson (the plain Level 3 solution AND a separately-maintained "Refactored Level 3" solution) - and this project has already directly experienced how costly maintaining two divergent code paths for the same lesson can be (see the Script/Visual parity cleanup ToDo item and GameMaker's Code-vs-Visual drift problem elsewhere in this document) - a second one, chasing a moving personal-style target lesson by lesson, is explicitly something he does not want repeated. Instead, this becomes a single dedicated final chapter/lecture, intended to fall OUTSIDE the 30-lesson cadence entirely - delivered as a final micro-lecture, something students take home at the end of the course to study and experiment with on their own, explicitly framed as an invitation for students to develop their OWN opinions, form, and style rather than simply inheriting Robert's. Not yet scoped: which specific refactorings to walk through, how long the micro-lecture runs, or whether it's written as a live-delivered lecture, a stand-alone reading, or both. *(Robert's decision, supersedes the earlier unscoped Level 4/refactoring idea; book-only, falls outside the 30-lesson structure)*
- ☐ Robert's own test-project reconciliation (GameMaker, Godot, and now Scratch) surfaced a real pattern in GameMaker and Godot, but a CORRECTED finding for Scratch specifically: Robert repeatedly moved shared logic (bullet/enemy collision detection/response, the shoot sound's trigger point) to whichever object he personally felt should own it while exploring his own eventual refactoring preferences - confirmed genuine, self-introduced drift in GameMaker and Godot (both already logged above). For Scratch, Claude initially (wrongly) told Robert his Enemy-owns-collision code was the same kind of drift and should be reconciled back to a Bullet-owns-collision design - this was an error, caught and corrected in the same session: Scratch's OWN canonical Key design switches hit-detection from the Bullet (Lesson 3-4) to the Enemy (Lesson 5 onward, and every lesson through 12) on purpose, documented directly in the Lesson 5 Key: "Scratch clones have no unique ID, so a Bullet clone can't identify exactly which Enemy clone it's touching from the Bullet's side - that's why the check moved to each Enemy clone's own script." So Robert's Scratch Bullet's lack of a touching-check was already correct and needs no change - only the shoot-sound placement (Bullet's own clone script vs. the Key's Player-side placement) remains an open, genuine style question for Scratch, not a correctness bug. Net effect: the cross-engine ownership-reassignment pattern below is real for GameMaker and Godot, but Scratch's Enemy-owns-collision is the engine's actual canonical design, not Robert's drift - worth keeping both facts straight for the eventual Rule 12 pass and the refactoring chapter. Separately, Robert's Scratch Bullet also spawns offset 100 pixels from the Player's exact position (y + 100) rather than exactly on it, done consistently the same way across his other engines' projects too - a minor, deliberate, and apparently cross-engine-consistent

deviation from the Keys' literal spawn position, not a mistake. Robert wants this one specifically noted and then removed when the book's canonical code is written, with the idea saved for the refactoring chapter rather than silently folded into the canonical design. *(Robert's explicit decision re: GameMaker/Godot ownership drift; Scratch correction found and caught within the same session after an initial wrong read by Claude)*

Notes & Commentary

Freeform observations worth remembering for when the book itself gets written - narrative angles, good anecdotes, structural ideas, and things that only became obvious by actually building the course. Seeded initially by pulling from the project's history up through the Unreal Lesson 2/3/4/13 rebuild and the Cross-Engine Consistency Reference document.

GML Visual: Set Instance Variable can't target a dynamically-created instance: Robert tested Lesson 5's GameMaker Visual color-switch instructions hands-on and found the originally-written steps didn't work: they called for dragging Create Instance's output into a "Set Instance Variable" block's Target field, but that block's targeting dropdown only actually offers Self, Other, or a specific object asset (which recolors every existing instance of that object, not just the newly spawned one) - there's no option to target an instance held in a variable at all. The fix he found and confirmed working: give Create Instance its own Target field (a new local variable, e.g. new_enemy), then use the generic "Assign Variable" block with a dot-notation Name field (new_enemy.image_blend) to reach that one specific instance. Corrected in Lesson 5's Handout and Key, and added as a named exception to Standards Rule 4 so a future contributor doesn't reintroduce the same mistake by over-applying "prefer the purpose-built block." Checked Lessons 3 and 13's similar-looking Set Instance Variable uses (targeting Self and Other, respectively, inside a Collision event) and confirmed both are legitimate - Self/Other are real, supported targets on that block; a Create-Instance-captured variable is the specific case it can't reach. No broader retrofit needed beyond Lesson 5 based on this check, but worth re-verifying if a similar spawn-and-tint pattern shows up in a not-yet-audited lesson.

Why 8 engines: The multi-engine structure (one shared weekly lesson, built out independently in Scratch, Godot, Unity, GameMaker, GDevelop, TIC-80, 001 Game Creator, and Unreal) is probably the book's core hook - it's an unusual structure for a game-programming book, most of which pick one engine. Worth an early chapter making the case for why explicitly: it demonstrates that the underlying concepts (loops, events, state, collision) are engine-agnostic, and lets a reader/student pick whichever engine matches how they think (text code, blocks, or pure visual wiring) without missing anything.

The Level 1/2/3 framework as the book's spine: MVP / Complete / Refined is used every single week for a small exercise, AND re-applied at the whole-semester level in Lesson 15 (MVP = Lesson 6, Complete = Lesson 11, Refined = Lesson 15). That fractal quality - the same rubric working at two very different zoom levels - is a strong structural device for the book itself, not just the course.

The Structure goal as a running architecture throughline: Every lesson's Structure goal is doing the same underlying job (separation of concerns) but the book should resist presenting it as one repeated checkbox - it genuinely escalates week to week: co-locate-by-trigger, then object-owns-its-own-behavior, then spawner-vs-behavior, then unrelated-counters-stay-unrelated, then single-reusable-entry-point (DRY), then consolidate-a-system-built-piecemeal. Full breakdown already captured in Cross_Engine_Consistency_Reference.docx - cite it rather than re-deriving this in the manuscript.

Rule 5 origin story is good anecdote material: The Unreal Lesson 2/3/4/13 rebuild this session is a genuinely good illustrative story for a sidebar or pull-quote: Unreal's content had quietly drifted into flat-color Materials where every other engine used the real supplied sprite, close enough to work but not actually consistent - which is exactly what prompted Robert to articulate the Engine Deviation rule (Standards Rule 5) explicitly. Good real example of "a rule written after the mistake it prevents," which is honest and relatable for readers building their own multi-platform anything.

Engine-specific gotchas as troubleshooting sidebar content: Several hard-won, engine-specific bugs surfaced this project are strong candidates for "common pitfalls" sidebars rather than being filed away as pure production trivia: Unreal's Blank template auto-possessing a flying DefaultPawn and eating movement input; the "multiply a color parameter by white is a no-op" trap that bit Godot, Unity, GameMaker, AND Unreal independently for the Lesson 13 hit-flash; GameMaker/001GC's generic "destroyed" hook firing for both a hit AND an escape in Lesson 7, requiring an actual code restructure. These read well as "here's a mistake you'll probably also make, and why it happens" content.

Production philosophy worth explaining to readers, not just following internally: The Self-Containment rule (no lesson sends a student hunting for an asset mid-class) and the Purpose-Built-Blocks rule (use the engine's specific named block over a generic one, in visual scripting) both read as genuine craft opinions a book reader would want stated outright, not just inferred from the finished lessons. Consider a short "how these lessons were actually built" chapter or appendix drawing on Lesson_Production_Standards.docx.

Source documents - don't duplicate, cite: Lesson_Production_Standards.docx and Cross_Engine_Consistency_Reference.docx (both in Lessons/) are the primary source material for anything about course philosophy, cross-engine rationale, or production conventions. This notes document should stay focused on book-specific decisions, open questions, and raw commentary - not a third copy of the same rules.

Rule 6 retrofit scope: forward-only vs. retrofit: Standards Rule 6 (SuperStretch tie-in, a Refinement and Style pass, and visual-scripting node-reuse guidance) is a standing requirement now, but it wasn't in force when Lessons 1-15 were built, so none of the 15 completed Handouts/Keys currently carry it. Two options: (1) retrofit - go back through all 15 lessons x up to 8 engines and add Rule 6 content to the live, already-shipped Handouts/Keys now; (2) forward-only - apply Rule 6 starting with Lesson 16, and write the Rule 6 content for Lessons 1-15 fresh at book-writing time instead, since the book chapter is a separate deliverable from the live-class Handout and can carry more reflective, after-the-fact content anyway. Leaning recommendation: forward-only. The SuperStretch tie-in is cheap to backfill later - most of that content already exists in the standalone SuperStretch docs and just needs pulling into context. The Refinement-and-Style and node-reuse content is genuinely new writing, at a scale (15 lessons x up to 8 engines) that would be a large side project for a live class with no real classroom benefit, since the students who used Lessons 1-15 have already moved past them. Better to write that material once, well, when the manuscript actually reaches those chapters, than twice - once retrofitted into a Handout already taught from, and again for the book. This is a recommendation, not a decision; Robert asked to scope this only and explicitly did not authorize starting the retrofit.

"Who's in charge" - a real Rule 6 case study, deliberately not retrofitted: Lesson 4's shipped Handout/Key (every engine) has one side of a collision - the Bullet in Unity/GameMaker/GDevelop/TIC-80, the Enemy in Godot - detect the hit and destroy BOTH objects. The cleaner alternative: Bullet and Enemy each detect their own hit independently and

destroy only themselves, so neither object needs to reach into and manipulate the other. Robert built his own project this way and confirmed the decision explicitly: keep the simpler, common beginner pattern in the live Lesson 4 content on purpose (functional progress first, for a student still getting collision detection working for the first time), and save the "who's in charge" framing - already introduced verbally in class, not yet written down anywhere - for a Rule 6 Refinement and Style section or a dedicated Sidebar later. Good, concrete, real-classroom example of Rule 6 in action: recognizing a legitimate improvement mid-lesson and consciously choosing NOT to interrupt functional progress with it. A companion example from the same conversation: Unity's Level 3 Sound idea already avoids a sound getting cut off by Destroy() via AudioSource.PlayClipAtPoint() (a detached, self-cleaning audio source) rather than Godot's hide+await+queue_free() dance - Robert had used a plain AudioSource as a child of the object instead, which gets torn down along with its parent even though Destroy()/queue_free() are deferred to end-of-frame, not instant. Same root cause as the collision-detection case (an object's behavior depending on something outside itself), same fix (self-contained ownership) - a nice paired example for the same Sidebar or Refinement section.

Diagram pilot: Lesson 5 flowchart + GameMaker node wireframe: First use of hand-drawn branded diagrams outside Lesson 2's original set, prompted by Robert noting he keeps re-explaining the same logic verbally in class. Two new diagrams, built as branded SVG (matching the existing Brand_Assets/Diagrams navy/gold UML style) and rendered to PNG via cairosvg: diagram_spawn_loop.png, an engine-agnostic activity diagram of the counter/repeat-check/create/increment shape behind every engine's Level 1 solution this lesson, embedded into all 8 engines' Handout AND Teacher's Key (right before the Level 1 steps/code); and diagram_gml_visual_wireframe.png, a GameMaker-only mock-up of the actual GML Visual block layout (Instance - Create Instance's Target field feeding Variable - Assign Variable's dot-notation Name field), embedded into GameMaker's Handout Level 3 Visual-Track section and Key altNotes section only. Scoped this way per Robert's explicit answers: diagrams live in both the Handout and the Key (not just one); this is a pilot on Lesson 5 only, not yet a retrofit of Lessons 1-4/6-15 and not yet a standing Standards Rule; and node-wireframe diagrams are GameMaker-only for now, since it's the only engine with students actively switching between Script and Visual tracks - Unreal was deliberately excluded from wireframe scope since Robert's two Unreal students recently moved to lighter-weight engines. See the new ToDo item for the retrofit/standing-rule decision, which is intentionally not made yet.

Diagram pilot promoted to Standards Rule 9 - full Lesson 5 extension: Robert decided the pilot's outcome directly: diagrams become a standing rule (Standards Rule 9), not a Lesson 5-only experiment, and he expanded the scope in the same request rather than narrowing it. The Activity Diagram is now standard in every lesson's Handout, Key, AND Slide Deck (previously Handout+Key only). Engine node diagrams now cover all three dual-track engines - GameMaker, Unity, and Unreal - reversing the earlier GameMaker-only wireframe scoping once Unreal's own student-enrollment status turned out not to matter for whether the diagram gets built. The Key always gets a node diagram (Level 1, plus a full combined-sequence diagram for Level 3 when the visual path is complex enough); the Handout always gets one for Level 1 specifically (added as an explicit follow-up note - previously Handout diagrams were novelty-gated only), plus an additional one when something genuinely unusual is introduced beyond Level 1 (the GameMaker back-door wireframe remains the model case for that half of the rule). Sidebars and Slides get diagrams opportunistically, when they'd clarify the text, not as a fixed requirement. Robert also flagged that a Slide Deck's diagram of a concept may deliberately not match an engine-specific document's diagram of the same idea, since engine-specific diagrams must authentically reflect that engine's actual visual paradigm (GameMaker's solid block-stack vs. Unity/Unreal's node-graph-with-pins) while Slides can simplify or generalize. Four new

diagrams were built to extend Lesson 5 under the new rule: diagram_gml_visual_level1.png and diagram_unity_visual_scripting_level1.png (Level 1-only node diagrams for the Handouts), and diagram_unreal_blueprint_level1.png plus diagram_unreal_blueprint_full_sequence.png (Unreal's first node diagrams in this course, for both Handout and Key). All four wired into build_lesson05_handouts_keys_checklist.js's data-driven diagram hooks; the generic diagram_spawn_loop.png was added to the Slide Deck's Loops section; and diagram_unreal_blueprint_full_sequence.png plus the existing diagram_gml_visual_wireframe.png were added to the Lesson 5 Sidebars' "Script vs. Visual Coding" topic to reinforce its existing Unreal-vs-GameMaker instance-reference discussion. See Standards Rule 9 for the formal rule text and the ToDo items above for the still-open past/future retrofit work.

Correction: Apply To solves the GML Visual targeting problem after all: Robert found, through his own hands-on testing in GameMaker, a cleaner solution to the exact problem the earlier "GML Visual: Set Instance Variable can't target a dynamically-created instance" note (above) documented - and it changes that note's conclusion. The earlier note's finding that Set Instance Variable's Action Scope dropdown can't reach a variable-held instance reference was correct as far as it went, but its fix (a generic Assign Variable block with a typed dot-notation Name field, new_enemy.image_blend) turns out not to be necessary. The actual solution: drag an "Instance - Apply To" block right after Create Instance, and attach whatever needs the scoped target - an If/Else block, in this lesson's case - to the RIGHT of Apply To, not below it, which is what nests it inside Apply To's scope rather than just sequencing after it. Apply To's own Action Scope pop-up has an Expression field at the top, separate from its Self/Other/object list, and typing the same variable Create Instance's Target field captured (new_enemy) into that field re-points what "Self" means for every block nested inside. That means the actual purpose-built block - Set Instance Colour, not the more generic Set Instance Variable - works perfectly once nested inside Apply To's scope, with no generic block or typed workaround needed at all. Corrected in Lesson 5's Handout and Key, in Standards Rule 4 (the "confirmed exception" is now a resolved case), and in the Sidebars' Script vs. Visual Coding topic, which had built an entire GameMaker-vs-Unreal comparison around GML Visual allegedly having "no socket" for an instance reference - Apply To's Expression field is exactly that socket, just reached through a scope-setting block and a pop-up field rather than a draggable wire on the canvas. Both diagrams (wireframe and full sequence) were redrawn to match. Worth a general lesson for future audits: a purpose-built block's own targeting dropdown looking limited doesn't necessarily mean the block can't be used - check whether a scope-setting block like Apply To (or an engine's equivalent) changes what's available before concluding a generic/typed fallback is the only option.

Ternary operator pulled from Lesson 5's GameMaker code; save it for Semester 2: Lesson 5's GameMaker Code path originally set image_blend using a one-line ternary, (i % 2 == 0) ? make_color_rgb(200, 25, 55) : make_color_rgb(230, 100, 130) - a pattern Robert genuinely likes and uses himself all the time, but judged a bit much to introduce to beginning students this early in the course. Rewritten as a plain if/else block instead, which is longer but reads unambiguously for a student who hasn't met conditional expressions as values yet. Robert's own suggestion: the Semester 2 capstone project - once students are comfortable with the fundamentals and are extending a more mature codebase - is a good, deliberate place to introduce the ternary operator formally, rather than leaving it to be picked up incidentally or never. Not yet scheduled to a specific Semester 2 lesson; see the paired ToDo item.

Demo-engine rotation drifted, then got re-planned as Standards Rule 10: Robert noticed Godot had been the live lecture-demo engine every single lesson from 3 through 14 (not just a

few weeks running, as he'd first suspected) and asked for a full per-lesson list, which surfaced the drift. A scan back through the project turned up real evidence that a rotation had genuinely been pre-planned, not just remembered inaccurately: the Syllabus, Engine & Platform Guide, and Engine Showcase docs all promise students an even rotation across engines/styles, and the 30-Lesson Master Schedule spreadsheet has an actual Featured Engine + Coding Style column for all 30 lessons modeling that balance. The promise was simply never enforced as a Standards Rule, so nothing stopped a lesson's "continue last week's demo project" framing from perpetuating whatever engine got picked once. Since Lessons 3-15's actual topics have diverged from the master schedule's original per-week order, the old schedule couldn't just be restored - a new rotation had to be designed against the lessons as actually built. Robert then narrowed the pool to only the engines/styles with active students (Scratch, GameMaker script, GameMaker visual, Unity C#, Godot), deferring full 8-engine/3-style balance to a ToDo item (above) rather than designing for engines nobody's using yet. Finalized rotation for Lessons 5-14, each choice appearing exactly twice, no back-to-back repeats, preserving a "same engine, two modes" GameMaker callback pairing (mirroring the original schedule's design intent): L5 GameMaker Visual (+ brief script contrast, ties to the Script-vs-Visual Sidebar), L6 Scratch, L7 Godot (unchanged), L8 Unity C#, L9 GameMaker script (callback to L5), L10 Godot (unchanged), L11 Unity C#, L12 GameMaker Visual (callback to L5), L13 Scratch, L14 GameMaker script (callback to L9). L15 stays demo-free (workshop/showcase day) per the original design. Every affected lesson's Plan (Featured Engine field), Teacher's Script (demo cues and talking points), and, for Lesson 5 only, Slide Deck speaker notes were rewritten to be technically accurate to their newly assigned engine - not just relabeled.

Enemy-vs-enemy self-collision: a real bug, found engine by engine, fixed quietly rather than taught yet: Robert hit a genuine runtime bug testing Lesson 5 Godot himself: no enemies appeared on screen or in the Remote Scene tree during play. Diagnosis (his own testing plus a review of the shared Hitbox/area_entered collision architecture from Lessons 3-4) found the real cause: Godot's Hitbox area_entered handler has no type-check on what triggered it, and at the original 40px-apart / 60px-spacing layout with 128x128px sprites, adjacent Enemies' hitboxes overlapped at spawn and mutually destroyed each other before the first frame was ever seen - confirmed directly by Robert reducing the loop count and watching enemies survive, then hearing the destroy sound fire once the count went back up. His chosen fix, deliberately the narrow one: widen spacing only (Vector2(150 + i * 200, 50)), explicitly declining the deeper architectural fix (a real Collision Layer/Mask setup, or a type-check in the handler) for now. Robert then reported Scratch has the identical problem - overlapping clones causing double-hits rather than full destruction - which he'd already independently fixed himself (enemy_i renamed to enemy_number, spacing formula changed to enemy_number * 100 - 300) before this was ever raised as a course-wide question. That prompted a full audit of the remaining six engines for the same generic-trigger-fires-on-any-overlap pattern: GameMaker is architecturally immune (its Collision Event is object-pair-specific, not generic); Unity (OnTriggerEnter2D + CompareTag), Unreal (Cast To BP_Bullet), GDevelop (named-pair collision condition), and TIC-80 (hand-coded rectangle-overlap math) are all safe by design for the same reason. Only 001 Game Creator was found vulnerable (its "Collided with actor" trigger is generic, with no actor-type check) - fixed the same way as Godot and Scratch, by strengthening the placement instructions in Lessons 5 and 14 ("at least a full enemy-width of empty gap") rather than restructuring the trigger logic. Robert was explicit about scope throughout: fix the practical problem silently, everywhere it exists, right now - but do not yet teach collision-type filtering (Bullet vs. Player vs. Enemy vs. Enemy) as a formal topic. A review of Lessons 6-15's actual content confirmed there was no orphaned teaching moment being skipped by this choice: true Player-vs-Enemy touch collision never appears anywhere in Semester 1 (only Bullet-vs-Enemy from Lesson 4, and an escape-based life-loss mechanic in Lesson 8), and Lesson 15's own

teacher script already frames collision-type filtering as Semester 2 territory. Good candidate for a Semester 2 sidebar or lesson-opener later: Robert's own framing for it, already used verbally in class per his message, was "Now our enemies are moving around, they can collide with each other AND the player. How do we fix this? What even is the right action we want for each type of collision?"

Cross-engine consistency promoted from habit to Standards Rule 12: While retrofitting Unity's Lesson 5 code to Rule 11 (meaningful variable names), a real naming conflict surfaced: Unity's EnemySpawner already has a public enemyCount field (the total number of enemies to spawn), so reusing that same name for the loop counter - the literal move Rule 11 alone would suggest, matching GameMaker and Godot's enemy_count - would shadow it, technically legal in C# but genuinely confusing. The loop counter was named enemyIndex instead, a distinct and still meaningful name, and the mismatch was flagged to Robert rather than resolved unilaterally. Robert's response went beyond just answering that one question: leave the enemyCount/enemyIndex mismatch alone for now, but treat the underlying question - keeping variable names AND code structure consistent across all 8 engines - as a fundamental precept of the course, not a nice-to-have. His reasoning, stated directly: he has personally observed students helping each other across engine and coding-style lines - a student on one engine comparing notes with a student on another, working through the same week's exercise - and the more common ground the actual names and structure share, the easier that peer help is. A student has to mentally translate "my enemy_count" to "their enemy_number" before they can even start comparing approaches; consistent naming and structure removes that translation tax entirely. This was true from the start of the multi-engine design but wasn't enforced strictly early on (Lessons 1-4 predate any naming rule at all, and Lesson 5's own Rule 11 fixes were applied engine-by-engine as each was caught rather than as one coordinated pass) - Robert is now retroactively naming it as something the course has valued implicitly all along, now that enough real classroom evidence exists to state it as a rule outright. Formalized as Standards Rule 12; a dedicated future cross-engine reconciliation pass (covering both naming and structure, across all lessons and engines) is tracked as a ToDo item above, deliberately scheduled later rather than forcing an immediate resolution of the Unity case that surfaced it.

Data Dictionary project launched to drive the Rule 12 consistency goal: Immediately after Rule 12 was formalized, Robert asked for a central, formal Data Dictionary - not just a ToDo describing the consistency goal, but an actual working reference naming every variable, function, and file the course creates: one common cross-engine concept name (e.g. "per-frame update"), paired with each engine's own actual name for that same concept where engines can't literally share one identifier. Undecided yet whether this will ever be shown to students, or stays a production-only tool - built either way, since it's useful internally regardless. Robert's first-choice format was a genuine Microsoft Access database with a data-entry Form and a Report, explicitly because he dislikes working in spreadsheets. Checked before building anything: Access's .accdb format (2007+) has no Linux-native authoring path at all - mdbtools, the standard Linux toolkit for Access files, doesn't even read/write the modern .accdb format (it's limited to the legacy Jet 3/4 .mdb format from Access 97/2000/2002), and more fundamentally, Access Forms and Reports are proprietary, VBA-hosted application objects with no file-level equivalent outside Access itself - no export format, no library, nothing short of the real Access application (Windows-only, not available in this sandboxed environment) can author a working Form or Report object, regardless of which .mdb/.accdb table format is targeted. Confirmed via both a direct tool check and a web search before ruling it out, rather than assuming. Robert had already pre-approved the fallback ("I'll live with an Excel spreadsheet but I'll pout about it"), so the dictionary is being built in Excel instead, with an explicit effort to make it feel less like a bare spreadsheet than the default - data-validation dropdowns, a proper per-engine reference layout,

and the closest practical approximation of the Form/Report feel Excel supports. Same conversation also raised the open case/style question (PascalCase vs. camelCase vs. snake_case) as a Rule 12 sub-decision, with Robert's stated leaning toward prioritizing cross-engine consistency in the chosen case style over each engine's own local idiom, explained via Sidebar wherever the two disagree - see the paired ToDo entry, not yet a final decision.

GameManager: a dedicated shared-state object, promoted from a Unity-only accident to Standards Rule 13: Reviewing Lesson 7 surfaced an asymmetry that had been sitting unnoticed since Lesson 6: Unity's Handout/Key already routed the win-counter (enemies_remaining) and win-flag (has_won) through a small dedicated GameManager object, while every other engine had that same shared state living wherever it happened to land - the spawner in some engines, a collision branch in others. This wasn't a deliberate Unity design choice being withheld from the rest of the course; it traced back to how Unity's Lesson 6 content happened to get written, and nobody had asked "why does only one engine do this" until Robert did. Once asked, the answer was obvious: a dedicated object whose only job is holding game-wide information is the better habit, full stop - Unity had simply stumbled into the right pattern first. Formalized as Standards Rule 13 (a dedicated manager object for any state more than one other object needs to read or change), and retrofitted into Lesson 6 across Scratch (a hidden GameManager sprite driven by broadcast/receive), Godot (a child GameManager Node reached via signal connection), GameMaker (obj_gamemanager, a placed instance), and Unreal (BP_GameManager, a placed Actor with a dispatcher binding) - Unity needed no change, having already modeled the target pattern. Good real-course illustration for the book of a rule discovered by noticing an inconsistency rather than designed up front - the opposite direction from Rule 5's origin story (a rule written after catching a mistake), this one was a rule written after catching an accidental correct answer nobody had generalized yet.

Lesson 7 compliance pass: GameManager routing, naming drift, and the full diagram/prose standard: With Rule 13 in place, Lesson 7 itself needed the same GameManager routing applied to its own win-tracking logic (all engines), plus a Teacher's Script callout telling students to go retrofit their own Lesson 6 project before starting Lesson 7's lab work, so the fix lands in every student's actual codebase rather than staying a documentation-only change. Auditing Lesson 7 against the full current Standards Rule set while making that change turned up two smaller drifts worth recording: Rule 11 (meaningful names) had regressed in a couple of spots - Scratch's enemies_left/enemy_i and Godot/GameMaker/Unity's bare speed field all got renamed to match the enemies_remaining/enemy_number/enemy_speed conventions established elsewhere - and Rule 9 (diagrams) hadn't been applied to Lesson 7 at all yet. Five new diagrams were built and wired into every Handout, Key, and the Slide Deck: a generic Activity Diagram contrasting the two ways an enemy can disappear (escape vs. hit, only one of which should decrement the counter - the exact bug this lesson's exercise centers on), a State Diagram for the enemy's two possible endings, and Level-1 node diagrams for the three dual-track engines (GameMaker GML Visual, Unity Visual Scripting, Unreal Blueprint). Unity's Visual Scripting altPath prose was also tightened afterward to name each node exactly as the diagram draws it (Update -> Translate -> Branch -> Destroy) rather than compressing the graph into a single dense sentence, matching the per-node granularity Lesson 5 set as the specificity bar for alt-path writing. Unreal's Blueprint path needed no equivalent prose tightening - it was already written at full specificity as the lesson's PRIMARY path (not an altPath), and the audit's original complaint about it (no diagram backing the prose) was resolved by the new diagram set.

Lesson 8 compliance pass: GameManager retrofit, naming fixes, and the full diagram/notes standard: Lesson 8 (the lives/lose-condition lesson - the mirror image of Lesson 6's win condition) got the same treatment as Lesson 7, this time against a slightly

different set of gaps. Rule 13: Lesson 8's lives counter and lose-flag (has_lost) were retrofitted onto each engine's GameManager object (mirroring exactly how enemies_remaining/has_won already route through it since the Lesson 6/7 retrofit), rather than living on the player or a collision handler directly - the same routing pattern, applied to the opposite condition. Rule 11/12: a genuine TIC-80 naming regression was caught and fixed (the bare j/e loop variables flagged as a known gap in the standing ToDo list finally got their Lesson 8 instance renamed to enemy_index/enemy, matching the Lesson 7 precedent), alongside a small Scratch naming drift back toward an earlier inconsistent variable name. Rule 7: GameMaker's Draw GUI altPath and Unity's Visual Scripting altPath both had field-value and node-name gaps tightened to full explicitness (exact Draw Text field values spelled out; Unity node names matched one-for-one to what the new diagram draws), closing the same category of gap Lesson 7's Rule 7 audit had already fixed elsewhere. Rule 9: a full diagram set was built and wired in - one generic Activity Diagram (the lose-condition logic, shared across every engine's Handout/Key/Slide Deck) plus three engine-specific Level 1 node diagrams (GameMaker GML Visual, Unity Visual Scripting, Unreal Blueprint), wired into every Handout and Key. This lesson surfaced a placement nuance not previously written down: in the Handout, the node diagram nests under the altPath for engines whose primary documented path is code (Unity, GameMaker), but sits as a top-level diagram for Unreal, whose primary path is Blueprint itself, not an alt; in the Key, the node diagram is always top-level for all three dual-track engines, since the Key presents the full combined solution rather than a split main/alt track. Rule 10/8: the Slide Deck had drifted to reference Godot in its live-demo materials list and Slide 9's content box, left over from before the Lesson 5-14 rotation was finalized (Lesson 8's assigned engine is Unity C#) - corrected to Unity/GameManager.RegisterEscape() throughout - and had zero addNotes() calls despite Robert's classroom now running Presenter View, so six speaker-notes entries were added (slides 5, 7, 9, 11, 13, 14 - matching the Lesson 6 reference pattern of skipping title/overview/divider slides), each carrying that slide's matching Teacher's Script section verbatim, including bracketed stage directions and timing checkpoints.

Lesson 9 compliance pass: score joins the GameManager, a Unity Rule 7 gap, and the full Rule 9 diagram/Rule 8 notes treatment: Lesson 9 ("On the Scoreboard" - score tracking plus a persistent on-screen HUD, introducing the Push vs. Poll display-update concept) got the same full audit-then-fix treatment as Lessons 7 and 8, requested proactively by Robert while he personally reviewed Lesson 8's materials rather than in response to a specific bug report. Rule 13 was the single biggest finding: Lesson 9 had fully regressed on the GameManager pattern across four engines, each in a different way - GameMaker was routing score through obj_spawner instead of obj_gamemanager; Scratch had the Enemy sprite's own script changing score directly instead of staying confined to GameManager's existing "when I receive [enemy hit]" broadcast handler; Godot had score and a new update_hud() function living on "the main scene's script" instead of GameManager's script, breaking the established get_node("../ScoreText")-from-GameManager pattern; and Unreal had Score and the HUD widget setup living on the Level Blueprint instead of BP_GameManager, the same Actor that already owns EnemiesRemaining and Lives. All four were corrected to route score (and, for Godot/Unreal, the new push-based update function) through the established manager object, cross-checked line-by-line against Lesson 8's already-compliant content as the reference baseline for each engine's correct object/function names rather than improvising a new pattern. Two smaller drifts were caught and fixed opportunistically along the way: a Rule 11/12 Scratch naming regression (enemies_left had crept back in, corrected to enemies_remaining), and a Rule 7 gap in Unity's Visual Scripting altPath (vague "a node" references tightened to exact node names and categories - Format String/String category, Add/Math-Scalar category - matching the specificity bar Lessons 5 and 8 established). Rule 9: a full diagram set was built - one generic Activity Diagram ("Score Joins the Hit Path - Then Push or Poll Takes Over,"

covering both the GameManager routing and the Push-vs-Poll architectural choice in one picture) plus three engine-specific Level 1 node diagrams (GameMaker GML Visual, Unity Visual Scripting, Unreal Blueprint) - and wired into every Handout and Key following the Lesson 8 precedent exactly: nested under the altPath in the Handout for GameMaker and Unity (whose primary path is code), top-level in the Handout for Unreal (whose primary path is Blueprint), and always top-level in the Key regardless of Handout placement, since the Key shows the combined solution. Rule 10/8: the Slide Deck's materials list and Slide 9 content box had drifted to reference Godot, left over from before the Lesson 5-14 rotation was finalized (Lesson 9's assigned engine is GameMaker script, a deliberate callback to Lesson 5's GameMaker Visual demo) - corrected to GameMaker/obj_gamemanager.score throughout - and had zero addNotes() calls, so six speaker-notes entries were added (slides 5, 7, 9, 11, 13, 14, the same pattern Lessons 6 and 8 established), each carrying that slide's matching Teacher's Script section verbatim.

GameMaker drag-and-drop uses instance variables (__dnd_lives, __dnd_score) - a correction found by testing: While reviewing Lesson 8 for GameMaker, Robert noticed that the Visual Scripting content relied on deprecated built-ins: the old global lives/score/health have been replaced by drag-and-drop's own instance-level variables, __dnd_lives and __dnd_score (dnd = drag and drop), and purpose-built blocks exist for them (Set Lives, If Lives, Draw Instance Lives, Set Score, Draw Instance Score), each with a Target drop-down that must point at the GameManager object. The earlier content had wrongly used generic Set Instance Variable blocks with Lives/Score as a supposed recognized built-in. The first proposed fix (keep a custom variable in code and leave the code and block paths separate) was rejected in favour of making the code path use the very same __dnd_ variables, so a student can mix typed code and blocks in one project. Lessons 8 and 9 were corrected on that basis; Lesson 8 also gained the missing sound assets (sfx_lifelost, sfx_gameover, sfx_start) so students never have to find their own. Good book anecdote: the author's own hands-on testing found what documentation-driven drafting missed, and the same session surfaced follow-up questions (dotted obj.variable vs. Target drop-down, Prefabs, asset-name prefixes) now tracked in the To-Do list.

GML Visual has no Else If block - a genuine engine-specific gotcha, not yet retrofitted: Robert's observation: GameMaker's GML Visual (drag-and-drop) toolbox appears to have no "Else If" node/block at all, unlike the code path's natural else if chain. The only way to express a third (or later) branch of a multi-way decision in Visual is to nest another If-style block one level deeper, inside the previous block's Else slot - so a three-way decision isn't three same-level blocks in a row, it's an If, whose Else contains another If, whose Else contains the final branch, and so on for each additional case. This is exactly the kind of engine-specific gotcha the project has already flagged as good troubleshooting-sidebar material elsewhere (see the earlier note on engine-specific gotchas as a candidate for a "common pitfalls" chapter): a student who's only ever written else if in text code will very reasonably go looking for an Else If block in GameMaker Visual, not find one, and assume the tool can't do what they want rather than realizing the shape is just different. Not yet checked against the shipped course content - see the matching To-Do item for which lessons (Lessons 8, 9, and 11 are the likely candidates, anywhere a lesson's logic needs to tell apart three or more outcomes - lives hitting zero vs. enemies hitting zero vs. neither, or a wave clearing vs. the whole round being won) actually have a three-or-more-way branch in their GML Visual content, and whether that content already nests correctly or was written as if a flat Else If existed.

Debugging 101 / Testing 101 / Testing-Debugging 201 - a companion course family, spitballed from this project's own bug history: Robert's idea, raised as a sideways thought rather than a scoped proposal: a new book/class, "Debugging 101," built around a code

repository pre-loaded with a variety of deliberately planted bugs for students to find and fix. The named bug categories he gave map with unusual precision onto bugs this very project actually produced and recorded: flat-out broken code that doesn't compile or run; a failed edge case (the Stalemate is exactly this - kills and escapes are each handled correctly in isolation, but the combination where some of each happen in one round was never considered); a new feature breaking an existing one (the dropped Level 3 sounds - a Rule 14 "complete method" regeneration for a later lesson silently deleted working Level 3 code from an earlier one); a mismatch between a design doc and the actual code (the WaveAnnouncementText/WaveText naming mismatch between a Handout's prose and its Teacher's Key); a broken refactoring (the GameMaker Visual targeting saga - a generic block was swapped for another generic block, then finally a purpose-built one, with two of the three attempts subtly wrong in a way only hands-on testing surfaced); and a misinterpreted feature (built to a literal reading of a spec, but not to what was actually intended - the enemy-vs-enemy self-collision bug is a plausible fit here too, where a generic "anything touched this" trigger was technically working as coded but not as meant). The parallel idea, "Testing 101": the inverse situation - an existing codebase plus a design doc, but no test suite at all, where the student's job is to write the tests. A possible synthesis, "Testing/Debugging 201": combine both so that writing a test case IS the mechanism that proves a fix actually worked, not just a separate skill taught alongside debugging - red test before the fix, green test after. None of this is scoped: no decision yet on audience (same Middle/High School homeschool audience as Video Game Creation, or a different one entirely), language/engine (the bug-category list reads as language-agnostic, but the actual repository would need a concrete stack), length (a standalone book, a short course, a unit folded into an existing one), or relationship to the main course (standalone companion vs. a Semester 2/Year 2 extension of Video Game Creation itself, given how directly its source material comes from this project). Recorded here purely so the idea isn't lost before it's had time to develop; revisit once Semester 1 of Video Game Creation is further along or complete.

Lesson 12 compliance pass: a real cross-engine sound-architecture conflict, resolved in favor of Option 1: Auditing Lesson 12's shipped materials (requested by Robert before review) surfaced a genuine design-decision conflict, not just a drift-style bug like Lessons 7-9's compliance passes found. The Handouts/Keys had Godot and Unity building a brand-new shared sound player this lesson - Godot's RoundSound AudioStreamPlayer2D swapping its .stream property between three clips, Unity's roundAudioSrc AudioSource with a brand-new GameManager.instance singleton calling PlayOneShot() - while Scratch and GameMaker simply added more one-shot sound calls to existing branches, no shared player at all. This wasn't a copy-paste typo: it directly contradicted the dedicated-per-event sound pattern Lessons 6 and 8 had already established and shipped in every engine (WinSound/LifeLostSound/GameOverSound as separate AudioStreamPlayer2D nodes in Godot; winSound/lifeLostSound/gameOverSound as separate Inspector-assigned AudioClip fields reached through the already-existing GameManager.instance in Unity). The Lesson 12 content also introduced a brand-new sfx_lose.wav asset, when sfx_gameover.wav already existed and served the identical game-over moment since Lesson 8 - an unnecessary and unexplained new asset, not a case where two legitimately different sounds were needed. Rather than resolve this unilaterally, Robert was asked directly: Option 1 was to retire the shared-player design and the new sfx_lose asset, folding Lesson 12's wave-clear sound into the existing per-event pattern (one more dedicated node/field, reusing sfx_gameover.wav for game-over exactly as Lesson 8 built it); Option 2 was to keep the shared-player design and treat it as this lesson's deliberate new teaching point (a legitimate, if different, architecture worth knowing), fixing only the naming/wrong-asset issues. Robert chose Option 1 for the live course - see the separate note below for Option 2's preserved reasoning for the book version. All four active engines' (Scratch, Godot, Unity, GameMaker) and all four paused engines' (GDevelop, TIC-80, 001

Game Creator, Unreal) Handouts and Keys were rewritten under Option 1: one new dedicated sound per engine for the wave-clear event (Godot's WaveSound node, Unity's waveClip field, GameMaker/Scratch/the rest's one new sfx_wave call), and sfx_gameover (and, where applicable, sfx_lifelost) made mandatory rather than optional, reusing the Lesson 8 assets rather than sfx_lose. The Prep Checklist, GDD sound-asset description, Plan, and Teacher's Script were all corrected to match (the Teacher's Script's featured-engine demo narrative, built entirely around the old shared-player idea, needed a full rewrite of its 0:15-0:35 segment). The Slide Deck's zero existing speaker notes (a separate Rule 8 gap, unrelated to the sound-architecture question) were filled in across all eight content slides while the deck was already being rebuilt for the wording fixes. The old Assets/sfx_lose.wav file in the Lesson 12 folder is now unused under Option 1 but was left in place rather than deleted unilaterally - flagged to Robert as an open cleanup decision.

Sound should have been its own dedicated lesson all along - a retrofit decision, and a sharper Level 1/Level 2 framing it surfaced: While getting ready to start on the just-fixed Lesson 12 materials, Robert had a realization he wished he'd had much earlier: Lesson 12 - "Hearing It Happen" - is fundamentally a dedicated sound lesson, and in hindsight the course would have been better served treating sound as a genuine, deliberately-scoped topic from the start, rather than scattering it as a series of small optional Level 3 'add a sound' stretch ideas across Lessons 2, 3, 4, 6, and 8 the way it actually happened. His stated regret, in his own words: introducing sound piecemeal in the earlier lessons 'was a mistake.' His proposed fix for the book version: strip the sound-related optional Level 3 content out of those earlier lessons entirely, and let Lesson 12 be where sound is taught properly, as its own concept with real attention - not as a backlog of previously-half-built features finally getting formalized, which is closer to what the live course's current Lesson 12 does. He left the door open to keeping some sound ideas as earlier-lesson stretch goals if that still makes sense on reflection, but the explicit design principle either way is to make Lesson 12 a genuine 'pull it all together' moment rather than a cleanup lesson. Importantly, this is scoped as book-only: the live course's Lessons 2-11 keep their existing optional-sound Level 3 content as shipped, and this session's Lesson 12 fix (reusing Lesson 6/8's established sounds and adding one new wave-clear cue) remains exactly right for the live course regardless of what the book eventually does differently in the earlier lessons.

A "Sound Manager" as the book-version Lesson 12 refactor target - resolving the scatter-vs-strip question: Robert, reviewing this session's Lesson 13-15 compliance work, raised a resolution to the open "strip sound out of early lessons" question above: "Sound Manager" (sometimes "Audio Manager") is a real, standard game-design pattern - a single dedicated object owning all audio playback (one or more reusable players, volume/mute controls, avoiding the same sound double-firing, sometimes music crossfading), the audio-specific instance of the general Service Locator pattern described in Robert Nystrom's Game Programming Patterns, and one of the most commonly taught "first singleton" concepts in Unity/Godot tutorials. His proposal: rather than stripping the scattered optional Level 3 sound stretch goals out of Lessons 2-8 for the book version (the earlier-recorded leaning), KEEP them - motivated students can experiment with sound early, nothing is lost - and instead make book-Lesson 12 the payoff lesson where whatever a student has scattered gets explicitly refactored into one formal SoundManager object/concept, rather than just adding more sounds to existing branches. Robert's stated reasoning: this gives Lesson 12 more pedagogical significance, turning it into a genuine 'extract into one reusable, centrally-owned thing' lesson - the same recurring teaching motif the course already uses elsewhere (GameManager under Rule 13; spawn_enemies() extracted into its own reusable function in Lessons 10-11; wave_cleared() consolidating two branches in the Lesson 10-11 Stalemate fix) - applied to audio instead of game state or

spawning. This also directly revives and sharpens the already-preserved Option 2 reasoning (the shared sound-player design retired from the live course in favor of Option 1, see the dedicated preservation note above): a formal SoundManager IS essentially Option 2's shared-player idea, now framed explicitly as a deliberate mid-course refactoring lesson (built on top of Lessons 2-8's scattered attempts) rather than a same-lesson architecture choice with no before-state to refactor from. Not decided or scoped - Robert said he'll review this discussion later - but recorded here since it meaningfully resolves the tension between the two previously-recorded, not-yet-reconciled notes on this topic (strip-sound-out vs. preserve-Option-2), and because of its direct kinship with the Rule 13 GameManager pattern and the course's other extract-into-a-manager lessons, it is likely the strongest candidate for how the book ultimately handles Lesson 12.

A sharper Level 1 vs. Level 2 framing, surfaced by the same conversation: Thinking through why scattering sound across early lessons felt wrong in hindsight led Robert to a broader, genuinely useful clarification of what Level 1 and Level 2 are actually FOR across the whole course - distinct from the existing 'Level 1/2/3 framework as the book's spine' note above, which covers the fractal MVP/Complete/Refined structure itself but doesn't address why a student would prefer one level over another. Robert's framing: Level 1 (Minimal/MVP) is specifically a classroom-pacing safety valve - its job is to give a student falling behind a fast, reachable floor so they don't fall so far behind the live class's cadence that they can no longer reasonably absorb the NEXT lesson, not to represent the course's actual intended destination for that week. Level 2 (Complete) is the real preferred target, especially for a self-paced or individually-studying reader (exactly the audience the book serves) - it's a more realistic, complete feature set, and it deliberately blends 'spoon-fed' instruction with 'figure it out yourself' challenge in a way Level 1 alone doesn't attempt to. This directly shapes the sound-consolidation decision above: Robert is fairly sure Level 2 goals should NOT include sound in the earlier lessons once sound is pulled out and consolidated into Lesson 12, since Level 2 is meant to be the realistic, preferred accomplishment for those lessons, and sound doesn't belong in that target picture until its own dedicated lesson actually earns it. Worth writing this Level 1-versus-Level 2 purpose distinction into the book's framework chapter explicitly, rather than leaving a reader to infer it from the numbering alone.

Lesson 15 Complete Codebase deliverable - scoped now, deliberately not built yet: Robert asked for a new Lesson 15 deliverable: a complete, finished codebase per engine (at least the active engines), handed to students at the end of Semester 1 so they have something concrete to check their own work against over the break and come back ready for Semester 2 - distinct from, but a direct realization of, the already-logged "Complete Script Document" Final Deliverable idea. He also raised a second, related complaint while discussing it: GameMaker Visual's node-based logic has been represented as prose description throughout this course so far ("I don't feel like a textual description has been the best approach"), and asked for an actually diagrammatic representation instead. Three scoping questions were asked and answered directly: engines (the active 5: Scratch, Godot, Unity C#, GameMaker Code, GameMaker Visual - the 4 paused engines excluded), depth (Level 3 complete code included wherever it exists, which means backfilling the known Rule 14 Level-3-prose-only gap first rather than settling for the Level 1/2 baseline - chosen specifically because that backfill work also feeds the book), and GML Visual representation (full node-flow diagrams per object, extending the existing Lesson 5/7/8/9/11 wireframe-diagram pilot to cover every object completely, not selective/trivial-skipped). The one firm instruction layered on top of all three answers: don't start building this yet. Robert was explicit that doing it now, before the remaining lessons have been reviewed, would likely be wasted effort - the reviewed lessons could still

change in ways that would require redoing the Complete Codebase work. Recorded here as a fully-scoped, ready-to-execute future task, deliberately not started.

"Doing It the Doers of Stuff Way" - why an ongoing Level 4 was abandoned in favor of one final chapter: Robert originally envisioned a fourth tier alongside Level 1/2/3, stretch goals, and Sidebars: something focused on refactoring and on introducing his own particular coding style, applied on an ongoing basis lesson by lesson the way Level 3 and Sidebars already are. Reflecting on this project's own real experience with lesson drift and his own code drift (see the GameMaker Code-vs-Visual parity ToDo, and the general pattern of naming/structure drift this project has repeatedly caught and fixed via Standards Rules 11-13), he concluded an ongoing Level 4 isn't workable in practice, for a specific, practical reason rather than a philosophical one: it forces a choice, every single lesson, between two options he considers both bad. Option A is making the refactor-to-his-style pass a mandatory part of every lesson going forward, which he explicitly does not want - it would slow lesson production and tie every future lesson to keeping a second code path current. Option B is maintaining two genuinely separate, divergent code bases per lesson - the plain Level 3 solution, and a separately-maintained "Refactored Level 3" - which this project has already experienced is a real, recurring maintenance cost (not a hypothetical one), given how much effort has already gone into catching and fixing drift between a single lesson's own Code and Visual paths within ONE engine, let alone an entirely separate refactored version of every lesson across all 8. His resolution: drop the ongoing-Level-4 idea entirely, and instead write a single, dedicated final chapter - "Doing It the Doers of Stuff Way" (a direct reference to his own studio name) - that does the refactoring discussion once, thoroughly, as its own standalone unit, explicitly outside the 30-lesson weekly cadence. Framed as a take-home capstone-adjacent lecture rather than lesson content proper, with an explicit invitation for students to develop their own coding opinions and style rather than simply adopting his - the chapter teaches the ACT and REASONING of refactoring and personal style as a transferable skill, using Robert's own style as the worked example, not as the prescribed answer.

Lesson 13 compliance pass: a severe multi-engine regression, a real Godot/Unity timing bug, and the first Activity Diagram added by student request: Robert asked for a full Standards Rule 1-15 audit and fix pass on Lesson 13, assuming it was "probably on the up and up" after Lesson 12's clean-up - instead, Lesson 13 turned out to carry the worst drift found in any compliance pass so far, likely because it was drafted before several of the GameManager-retrofit and naming conventions established by Lessons 7-12 existed. GameMaker's Level 1 code still called obj_spawner.enemies_remaining/obj_spawner.spawn_wave() throughout, months after Lesson 9's compliance pass had already routed enemies_remaining through obj_gamemanager and Lesson 10's spawn_enemies() rename; TIC-80 had the matching stale spawn_wave() call; GameMaker's altPath still used a generic Set Instance Variable block on sprite_index instead of the purpose-built Set Sprite block Rule 4 requires (the same category of gap the __dnd_lives/__dnd_score correction caught in Lessons 8-9). Scratch's broadcast naming had drifted to Title Case ([Enemy Hit] instead of the shipped convention's lowercase [enemy hit]) during the rewrite and was caught and corrected before shipping. More seriously, both Unity and Godot's Level 1 code had a genuine timing bug baked into the Key itself: GameManager.RegisterKill() (Unity) and destroyed.emit() (Godot) were sequenced AFTER the 0.1-second hit-reaction delay instead of before it, meaning score/wave/win-check would lag visibly behind every hit - directly contradicting this lesson's own central teaching point (the delay should affect ONLY how the enemy's removal looks, never when game state changes). Both were corrected to fire immediately, with only the enemy's actual destruction (Destroy(gameObject) / queue_free()) staying behind the delay. Rule 10: the Slide Deck's live-demo materials list still named Godot instead of Scratch (Lesson 13's actual assigned rotation engine). Rule 8: the deck had zero speaker notes; five were added (slides 5, 7, 9, 11, 13)

carrying the Teacher's Script's matching timed sections verbatim. Rule 9: Lesson 13 had no diagrams at all, and Robert specifically asked for an Activity Diagram after students asked about them directly in class - a new generic Activity Diagram ("A Hit Happens - What Runs Now, What Waits," the first diagram this project has built in direct response to a student question rather than an author-driven audit finding) plus a GameMaker Visual Level 1 node diagram (obj_bullet's revised Collision event plus obj_enemy's new Alarm 0 event, the first diagram to show a GameMaker Visual object gaining a brand-new event rather than just new blocks in an existing one) were built and wired into every engine's Handout and Key, GameMaker's Handout and Key additionally getting the node diagram under its altPath/top-level per the established Lesson 8 placement convention. All affected build scripts (Handouts/Keys/Checklist, Slide Deck) were rebuilt, converted to PDF, and verified via pdftotext that no stale obj_spawner/spawn_wave/Title-Case-broadcast text remained and that both engines' corrected event ordering appears in the shipped text, before shipping to the Lesson 13 folder.

Lesson 14 compliance pass: a stale spawn-routine name drifted into five engines, a genuine Godot/GameMaker ordering bug, and a real Rule 12 discovery (TIC-80 legitimately calls it spawn_wave()): Robert granted standing authorization to audit and fix Lessons 14 and 15 to the Lesson 13 standard while away from the keyboard, to be reviewed together as completed work on his return. Lesson 14's audit found the most widespread single naming bug of any compliance pass: the Handout/Key content called the enemy-spawning routine spawn_wave() in Scratch, Godot, GameMaker, and (correctly) TIC-80 alike - but Lessons 10-13 had already established spawn_enemies() as the real function name in Godot, GameMaker, and Scratch (Unity's SpawnEnemies() and GDevelop/001/Unreal's event-based names were already correct and needed no change). Scratch additionally used enemies_left instead of the shipped enemies_remaining, and - in a near-miss caught only by cross-checking Lesson 11/12's actual canonical broadcast casing - used [spawn wave] instead of the real lowercase [spawn enemies] broadcast name. Investigating TIC-80 specifically to confirm its correct name turned up a genuine, useful discovery: TIC-80 is NOT drifted at all, and actually does call its routine spawn_wave() - a real, pre-existing Rule 12 cross-engine naming inconsistency (TIC-80 says spawn_wave, four other engines say spawn_enemies, Unity says SpawnEnemies) that was sitting undocumented until this audit surfaced it by requiring a definitive per-engine check. Beyond naming, two real structural bugs were caught and fixed: Godot's start_game() resets enemies_remaining BEFORE wave = 1 runs (the opposite order from GameMaker's start_game(), which resets wave before enemies_remaining) - Lesson 14's new wave-dependent formula would have silently used the previous game's leftover wave number unless the Handout/Key explicitly flagged the ordering dependency, which they now do; and GameMaker's Level 1 content had briefly folded the enemies_remaining reset INSIDE spawn_enemies() itself (changing the function's long-established contract and collapsing the 'two places must agree' lesson into one place for GameMaker only) before being corrected to keep spawn_enemies() creating instances only, with enemies_remaining staying split across start_game() and obj_bullet's win-check else branch exactly as Lesson 11 originally built it. The Slide Deck's Rule 10 demo-engine reference (and its spawn_wave()/speed-already-exists content, both leftover from before this lesson's actual GameMaker-featured, GameMaker-needs-a-new-speed-variable design) was corrected to GameMaker throughout, the Teacher's Script's matching spawn_wave() reference was fixed, and five speaker-notes entries were added (Rule 8, zero existing). Rule 9: a new generic Activity Diagram ("One Formula, Two Places - They Must Always Agree") and a GameMaker Visual node diagram (spanning all three places that change: spawn_enemies(), start_game(), and obj_bullet's win-check else branch) were built and wired into every engine's Handout and Key. While fixing this lesson, two bugs were also found and corrected in the just-shipped Lesson 13: a wrongly Title-Cased Scratch broadcast name ([Spawn Enemies] instead of the real lowercase [spawn enemies], introduced

during the Lesson 13 fix pass itself) and a TIC-80 regression where that same Lesson 13 pass had incorrectly renamed TIC-80's correct spawn_wave() to spawn_enemies() and used the wrong sfx slot number (sfx(4), Lesson 8's life-lost sound, instead of sfx(6), the actual wave-clear slot established in Lesson 12) - both corrected and reshipped before starting Lesson 14's own fixes.

Lesson 15 compliance pass: a stale obj_spawner reference, an abandoned Option-2 Unity field name, and the project's first process-only Activity Diagram: Lesson 15 ("Ready to Show," the Semester 1 capstone workshop-and-presentation day) got the same compliance treatment as Lessons 13 and 14, completing the three-lesson review Robert authorized while away from the keyboard. Structurally this lesson is unlike every other audit so far - it introduces no new mechanic, has no featured lecture-demo engine, and its Handouts/Keys are a self-assessment checklist and a 'known remaining quirks' troubleshooting reference rather than a worked Level 1 solution, so most of Rules 4/7/14 (which govern new code content) simply did not apply. Two genuine drift bugs were still found and fixed: GameMaker's quirks/diagnostics referenced a missing obj_spawner. prefix as a common last-mile bug, when the actual shared-state owner has been obj_gamemanager since the Lesson 9 compliance pass - corrected in both the Handout's engine notes and the Key's quirks list. Unity's quirks referenced roundAudioSrc as an example of a commonly-unassigned Inspector field - roundAudioSrc was the shared-sound-player field name from Lesson 12's abandoned Option 2 design (see the dedicated Option 2 preservation note), retired in favor of per-event fields when Robert chose Option 1; corrected to reference waveClip, one of the actual current per-event fields. Rule 10 did not apply (no lecture-demo engine this week, confirmed against the Plan's own "No lecture demo this week" format note), but the Slide Deck's two milestone-framing sections (Welcome Back, Milestone Moment) were each separately tagged 10 minutes despite the Teacher's Script and Plan agenda treating them as one combined 10-minute block - corrected to 5 minutes each so the on-screen pacing matches the actual schedule. Rule 8: zero existing speaker notes; five were added (slides 5, 7, 9, 11, 14), split across the Teacher's Script's five timed sections by best content match rather than rigid slide-boundary alignment, since the deck's own section structure (adding a dedicated "Milestone Moment" section) doesn't map one-to-one onto the script's five written sections. Rule 9: although this lesson teaches no new mechanic, the Standards call for an Activity Diagram in every lesson, so a new generic diagram - "Today's Workflow - Checklist, Polish, Present" - was built depicting the day's actual process (work the checklist, loop back through "Fix It First" if anything's broken, then tune/polish, then present) rather than any code logic, the first diagram in this project built around a class-day workflow instead of a programming concept, and wired into every engine's Handout and Key (no engine-specific node diagram applies, since no engine has new Level 1 code this week).

Lesson 12's Option 2 (shared sound-player design) - preserved reasoning for the book version: Recorded per Robert's explicit instruction, separate from the Option 1 implementation note above: even though Option 1 was chosen for the live course, the shared-sound-player idea the Lesson 12 content had already built (Godot's RoundSound swapping .stream between win/wave/lose clips; Unity's roundAudioSrc with PlayOneShot() layering three clips through one AudioSource) is a genuinely different, and genuinely teachable, architecture - not simply a mistake to discard. The case for reviving it in the book version: (1) it's a real-world pattern working game developers actually use - many shipped games route all short, mutually-exclusive one-shot UI/feedback sounds through a single pooled or shared AudioSource/AudioStreamPlayer rather than one dedicated node per sound, specifically to avoid the node/component count growing linearly with every new sound effect a game ever adds; (2) it's a natural, motivated moment to introduce the concept of a resource (an AudioClip/stream reference) being swapped at runtime rather than fixed at design time, which

dedicated per-event nodes never require a student to think about; (3) PlayOneShot() (Unity) and a stream-swap-then-play() (Godot) are each worth teaching in their own right as engine-idiomatic answers to 'how do I play one of several interchangeable sounds from one player,' independent of whether this specific lesson's three sounds (win/wave/lose) technically need it; (4) it would make a clean point of comparison against Scratch and GameMaker's simpler per-branch sound calls, turning the cross-engine contrast itself into a teaching moment ('two valid architectures for the same problem, here's the tradeoff') rather than something to quietly reconcile away. The case against reviving it, for balance: it does contradict the dedicated-per-event pattern Lessons 6 and 8 spent two lessons establishing, so introducing it here reads as a late architecture change rather than a planned progression unless the book's Lesson 6/8 chapters are rewritten to set up the contrast deliberately in advance; it also adds genuine new vocabulary (resource swapping, PlayOneShot's layering semantics vs. .Play()'s replace semantics) in a lesson Robert designed to be lighter than most, which may not fit Lesson 12's intended pacing even in the book. If revived, the clean approach would likely be: keep Lessons 6/8 exactly as the live course now has them (dedicated per-event nodes, Option 1 style), then introduce the shared-player idea in Lesson 12 as a DELIBERATE architectural pivot with its own named rationale ('your game now has four round-ending sounds - here's a different way to organize that'), rather than retroactively implying Lessons 6/8 should have used it from the start. Not decided; purely preserved reasoning for whoever revisits this at book-writing time, per Robert's explicit request to keep it rather than let Option 1's implementation erase the alternative's thinking.

Robert's own Unity project reconciliation: a fourth engine with the ownership-reassignment pattern, plus two real incidental bugs: Following the same process already used for GameMaker, Godot, and Scratch, Robert reconciled his own Unity Lesson 12-level project against canonical, starting with Bullet.cs and Enemy.cs. His Bullet.cs matched canonical's flash-on-spawn visual effect (color swap + scale pulse via a flashTimer countdown, structurally identical to Lesson 3's Complete Level 3 solution) but used Color.orange instead of canonical's Color.yellow - confirmed by Robert to compile and work fine (not flagged further). More significantly, Robert's project had moved bullet/enemy collision ownership onto Enemy.cs (OnTriggerEnter2D, the hit sound, and GameManager.RegisterKill() all living there) rather than Bullet.cs, matching the exact same deliberate-exploration pattern already tracked above for GameMaker and Godot - now a fourth confirmed data point, and the first one found in Unity specifically. Separately, Robert's Bullet.cs also played its own fire/shoot sound in Start() via a private AudioSource field, which he confirmed is a deliberate ownership choice (not a miss) and asked to be preserved as-is rather than reconciled to canonical's Player-owns-the-shot-sound placement - tracked as raw material for the 'Doing It the Doers of Stuff Way' chapter rather than fixed. Both files were rewritten to canonical for the current course version per Robert's explicit instruction: Bullet.cs regained OnTriggerEnter2D (checking the Enemy tag, playing hitSound via AudioSource.PlayClipAtPoint, calling GameManager.RegisterKill(), destroying both objects) and kept the flash effect (reverted to canonical's Color.yellow) and the fire-sound-in-Start() exactly as Robert wanted it kept; Enemy.cs was stripped back down to movement, the off-screen escape check, and GameManager.RegisterEscape() only. Two real, incidental bugs were caught along the way and flagged to Robert (not yet confirmed fixed in his project): Enemy.cs's OnTriggerEnter2D was checking other.CompareTag("Player") instead of "Bullet" - meaning hit detection likely wasn't registering via that path at all, since nothing in the game's design has the Player touching an Enemy register as a hit; and Enemy.cs carried a stray, unused using UnityEditor.UI; at the top, which compiles fine in the Editor but would break a standalone player build - Robert confirmed it was already greyed out as unused and wasn't sure how it got there. Robert separately confirmed his actual Bullet prefab is tagged "Player" rather than "Bullet", explaining the CompareTag("Player") bug directly - likely a copy-paste-from-Player-prefab origin

rather than a code mistake, with the fix being a Tag dropdown change on the prefab itself (and a quick check that the Enemy prefab is actually tagged "Enemy" too), not a code change. Also worth flagging separately (not part of this reconciliation, not yet fixed in the shipped course materials): the shipped Lesson 13 Unity content incorrectly describes hit-detection as having always lived on 'the Enemy prefab's script,' when canonical Lessons 4-12 confirm it has always been Bullet.cs - missed during Lesson 13's own compliance pass, tracked here as a found-but-not-yet-fixed continuity bug in the live course materials.

CORRECTION: the delivered Bullet.cs rewrite dropped the fire sound it was supposed to keep: The note above states the fire/shoot-sound-on-Bullet placement was 'kept as-is' per Robert's explicit instruction - that was wrong. Claude said it would keep it, then actually delivered a rewritten Bullet.cs with the audioSrc field and its Start()-time .Play() call removed entirely, directly contradicting Robert's stated intent and the record Claude had just written. Robert caught this by its real symptom (fire and hit sounds both gone) rather than by the code review itself. Hit sound was a separate, smaller issue layered on top: hitSound is a new field on Bullet (moved from Enemy's old enemyHit field as part of the collision-ownership reconciliation), so even with correct code the Inspector slot for it was never assigned on Robert's Bullet prefab - the old enemyHit assignment on Enemy doesn't carry over to a differently-named field on a different object. Both fixed: audioSrc and its Start() call restored to Bullet.cs, and Robert still needs to drag his hit-sound clip onto the new hitSound slot. Worth remembering for future reconciliation passes: double-check that delivered code actually matches what was promised in the same message, not just that the promise sounds right.

SECOND CORRECTION: Robert's hit sound wasn't actually missing, and his own fire-sound-ownership call changed too: Following on directly from the correction above. Robert re-tested and found the hit sound was never actually gone - it was present but extremely quiet (had to turn his laptop volume up to hear it) - and separately decided, after thinking about it more, that he'd rather the fire sound live in the canonical place (Player's own script) after all, so cross-engine logic stays consistent; his original ownership argument is saved for the refactoring chapter instead of kept live now. Bullet.cs's fire-sound field and Start() call were removed again (this time deliberately, matching canonical, not by mistake). The quiet-hit-sound symptom turned out to be real and diagnosable: Claude's first suggested fix (bump the PlayClipAtPoint volumeScale argument) was wrong - Robert correctly identified that volumeScale is clamped 0-1, a percentage of the clip's own volume, not a boost, so it can never counteract 3D distance rolloff. The actual, confirmed-working fix is calling PlayClipAtPoint at Camera.main.transform.position instead of the hit location - which is exactly the pattern GameManager's own round-ending sounds already used, and which turned out to be a genuine, previously undocumented bug in the shipped Lesson 4 Unity materials (tracked separately as its own ToDo and Notes entry) as well as a real regression introduced into Lesson 13's HandleHit() Coroutine rewrite (hitSound was dropped from that function entirely - fixed and reshipped). Robert's own Bullet.cs's hitSound call was corrected to the camera-position pattern too. The canonical Player-side fire sound is still pending - Robert's actual fire/shoot script hasn't been reconciled yet, since PlayerMovement.cs (reviewed and fixed earlier) turned out to contain no shooting logic at all.

Robert's own Unity GameManager.cs reconciliation: one genuine gap (missing waveClip), one legitimate addition worth keeping: Continuing the file-by-file Unity reconciliation after Bullet.cs/Enemy.cs, Robert's GameManager.cs turned out to already match canonical closely at the Level 3 Stalemate-fixed architecture - Update()'s new-Input-System restart check, ResetGame(), the EnemySpawner field, and RegisterKill()/RegisterEscape() all lined up with the shipped Lesson 10/11 Key almost exactly (matching formatting/commented-

out-old-code noise aside). One genuine gap: Lesson 12's entire addition, the waveClip AudioClip field and its AudioSource.PlayClipAtPoint() call on wave-advance, was missing outright. Because Robert's WaveCleared() (the Lesson 11 Stalemate-fix architecture) already exists, ahead of where Lesson 12's own Level 1 instructions assume a project is (they show the call inline in RegisterKill()'s nested if/else, since Level 1 doesn't have WaveCleared() yet), the call was added inside WaveCleared()'s else branch instead, next to winSound - the correct integration point for a project already at this level of accomplishment, not a literal copy of the Level 1 snippet. Separately, Robert's Start() plays a gameStartSound via AudioSource.PlayClipAtPoint() - not canonical GameManager code, but not drift either: sfx_start.wav was shipped all the way back in Lesson 6's Assets folder explicitly as a general-use stinger, and Robert is using it exactly as intended. Recommended he keep it. The reconciled file also dropped a stray commented-out using Unity.VisualScripting.Antlr3.Runtime.Collections; at the top and several dead commented-out code blocks (pre-Stalemate-fix RegisterKill()/RegisterEscape() logic Robert had left in place rather than deleted) - cosmetic cleanup, not an architecture question.

HUD.cs as a live-classroom debugging technique - worth teaching on purpose, not just confessing to: Reconciling Robert's HUD.cs (otherwise an exact match for canonical's scoreText/livesText/waveText) turned up one extra field: a Text displaying GameManager.gameActive.ToString() as raw True/False. Robert confirmed it was a debug leftover - but specifically one he'd used deliberately in class to demonstrate a technique: when printing to the console (or another normal debugging method) is inconvenient to show a room of students, temporarily wiring a bit of internal state straight onto the visible HUD is a legitimate, fast way to make it observable to everyone at once. Worth promoting from 'thing Robert happened to leave in' to an actual named technique/sidebar - 'abusing the HUD for debugging' - since it's a real, reusable classroom move, not just an artifact of one lesson's build. The field itself was still removed from Robert's reconciled HUD.cs as debug-only, not meant to ship in the finished game.

Robert's own Unity PlayerMovement.cs reconciliation: two real behavior gaps from splitting one combined check into four: The last file in the Unity Bullet/Enemy/GameManager/EnemySpawner/HUD/PlayerMovement reconciliation sequence. Movement, the screen clamp, directional sprites, and the footstep blink/sound were all present and matched canonical's Lesson 2 Complete Level 3 solution in spirit, but Robert's version checked each arrow key in its own separate if block (each calling a shared ImWalkingNow() helper) rather than canonical's structure of computing h/v first, then doing blink/sound/sprite exactly once inside a single combined moving check. Two real, if subtle, behavioral differences fell out of that structural choice: holding two direction keys at once (diagonal movement) calls ImWalkingNow() twice in the same frame, and blinkTime += Time.deltaTime isn't guarded against that, so the blink rate quietly speeds up only during diagonal movement; and there was no walkingBeep.Stop() anywhere - canonical explicitly stops the AudioSource when idle, but Robert's version only ever calls .Play(), so if that AudioClip loops (likely, for a continuous footstep beep), the sound keeps looping forever after the player stops moving. Reconciled back to canonical's single-combined-check structure, which fixes both. Two legitimate keepers, not reconciled away: named xBoundary/yBoundary fields (8f/4.5f) instead of canonical's hardcoded literals - a genuine readability improvement; and sr wired via a [SerializeField] private Inspector reference instead of canonical's GetComponent<SpriteRenderer>() call in Start() - a valid alternative wiring style, same SerializeField-vs-public question already tracked as its own ToDo. Robert still needs to re-drag the Inspector field references and playtest before confirming this version works in his actual project - not yet verified at the keyboard.

Unity reconciliation closed out: Player.cs's fire sound, the last piece: Robert had missed his own Player.cs in the file list - once found, it turned out to already match canonical almost exactly (fireCooldown decrement, space-key check, Instantiate, 0.2s cooldown reset). One pre-existing, already-tracked deviation: Player.cs spawns the bullet offset 0.7 units above the Player's own position (bulletYOffset, a named constant - nicer than the magic numbers used in his other engines' projects) rather than exactly on it, the same cross-engine-consistent deliberate offset already logged above for Scratch/GameMaker/Godot - now confirmed as a FOURTH engine with the same personal habit. The only real gap was the fire sound itself, entirely missing (no AudioSource field at all) - added as a SerializeField AudioSource, with audioSrc.Play() called right next to Instantiate(), exactly where canonical puts it. This one doesn't need the Camera.main.transform.position treatment hitSound needed - a persistent, Inspector-added AudioSource defaults to 2D spatial blend, so it was never subject to PlayClipAtPoint()'s 3D distance-rolloff problem in the first place. This closes out the full Unity reconciliation sequence: Bullet.cs, Enemy.cs, GameManager.cs, EnemySpawner.cs, HUD.cs, PlayerMovement.cs, and finally Player.cs, all now matching canonical per Robert's own assessment ('mostly' - the few remaining differences being his own confirmed, deliberate keepers: the bullet-spawn offset and the [SerializeField] private vs. public fields question, both already tracked above).

Robert's own marker, for himself: Lesson 12 is the Git checkpoint where his experimental Unity code was deliberately reconciled back to [mostly] canonical: Robert asked for this recorded explicitly, mainly as a personal note rather than course content: the reconciliation pass that just finished (Bullet/Enemy/GameManager/EnemySpawner/HUD/PlayerMovement/Player) is the point where his own Unity project's accumulated experimentation - the bullet/enemy collision-ownership exploration, the fire-sound-placement question, the gameActive HUD debugging technique, the SerializeField/public style question, assorted dead/commented-out code - was deliberately stripped back out and the course code returned to [mostly] canonical, specifically so he can find his own pre-reconciliation experimental code again later in Git if he wants to revisit any of it (for the refactoring chapter, or just out of curiosity) without having to reconstruct it from memory. Treat this entry as a locator/bookmark, not a course-content item - the actual content of what was reconciled is already captured in the matching Notes & Commentary entries above.

Revision Note

First entry, written the same week as the Unreal Lesson 2/3/4/13 rebuild and the first pass of Cross_Engine_Consistency_Reference.docx. Updated the same week to add Standards Rule 6 (SuperStretch/Refinement and Style/visual-scripting reuse) and scope, but not begin, the question of retrofitting it into Lessons 1-15. Updated again to add Standards Rule 13 (dedicated manager object for shared state), the Lesson 6 GameManager retrofit, and Lesson 7's full compliance pass (GameManager routing, Rule 11 naming-drift fixes, the new Rule 9 diagram set, and Unity Visual Scripting prose tightening). Updated again to add a GameManager mini-lecture to Lesson 7's live slide deck (placed at the start of the lab section, separate from the main lecture) plus the paired ToDo to relocate that discussion to Lecture 06 once the book manuscript reaches it. Updated again to add Lesson 8's own full compliance pass (the lives/has_lost GameManager retrofit, Rule 11/12 TIC-80 and Scratch naming fixes, Rule 7 visual-path explicitness fixes, a new Rule 9 diagram set with a newly-documented Handout-vs-Key placement convention for the three dual-track engines, and the Rule 10/Rule 8 slide deck fixes correcting a Godot-leftover demo reference and adding its first six speaker-notes entries), and to narrow the Rule 7/8/9/13 retrofit ToDo items' remaining scope accordingly. Updated

again to add Lesson 9's own full compliance pass (a four-engine Rule 13 GameManager retrofit for score - GameMaker, Scratch, Godot, and Unreal - plus a Rule 11/12 Scratch naming fix, a Rule 7 Unity Visual Scripting explicitness fix, a new Rule 9 diagram set reusing the established Handout-vs-Key placement convention, and the Rule 10/Rule 8 slide deck fixes correcting a Godot-leftover demo reference and adding its six speaker-notes entries), and to narrow the Rule 7/8/9/13 retrofit ToDo items' remaining scope to Lessons 10-15 accordingly. Updated again to record the GameMaker __dnd_lives/__dnd_score correction (Lessons 8-9) and five follow-up To-Do items (Lessons 10-15 review, obj.variable vs. Target drop-down, an If Lives comparison sidebar, GameMaker Prefabs, and dropping asset-name prefixes). Updated again to record the Unity static-members items, the cross-engine name normalization To-Do and the canonical bug-hunt decision, plus the Lesson 10 compliance pass and the cross-engine design-drift To-Do (HUD object, prefabs). Updated again at the Lesson 11 review to point to the new Reflex Habits design-notes document and to record the name "the Stalemate" for the win/lose gap bug and the Lesson 11 compliance pass (active engines only). Updated again to record a spitballed new companion course family - Debugging 101, Testing 101, and a combined Testing/Debugging 201 - raised by Robert as a sideways idea, not yet scoped. Updated again to record the GML Visual "no Else If block" gotcha (not yet audited against shipped content) and to extend Standards Rule 12's scope to cross-engine pop-up/HUD text presentation (centering, placement, size), with reasonable shared defaults proposed but not yet decided or audited. Updated again to record Lesson 12's compliance pass and its genuine sound-architecture conflict (a new shared sound-player design in Godot/Unity contradicting the Lesson 6/8 dedicated-per-event pattern, resolved for the live course as Option 1 - retire the shared player, reuse Lesson 8's sfx_gameover.wav instead of a new sfx_lose.wav asset), plus a separate note preserving Option 2's reasoning (the shared-player design) for reconsideration at book-writing time. Updated again to record Robert's book-retrofit decision to pull all scattered optional sound content out of Lessons 2-8 and consolidate it into a genuinely dedicated Lesson 12 (live course unaffected), plus the sharper Level 1-versus-Level 2 purpose framing it surfaced - Level 1 as a classroom-pacing safety valve, Level 2 as the real preferred target especially for self-paced study - with a leaning that Level 2 goals should exclude sound in the earlier lessons once consolidated. Updated again to record Robert's decision to abandon an ongoing Level 4 (refactoring/personal-style) concept in favor of a single dedicated final chapter, "Doing It the Doers of Stuff Way," delivered outside the 30-lesson cadence as a take-home capstone-adjacent lecture. Updated again to fully scope a new Lesson 15 Complete Codebase deliverable (active 5 engines, Level 3 code included via a Rule 14 backfill, GameMaker Visual represented as full per-object node-flow diagrams rather than prose) - explicitly deferred, not started, until the remaining lessons are reviewed. Updated again to record a GameMaker Lesson 12/wave_cleared() reconciliation note (Level 1 vs. the Lesson 10/11 Level 3 Stalemate-fix architecture), surfaced while Robert reconciled his own drifted GameMaker Script project - explicitly left as-is for now, earmarked for either a later explicit refactor or the planned final refactoring chapter. Updated again to record six candidate refactorings Robert spotted while bringing his own GameMaker project back into compliance (enemy-count self-reporting on Create, a named enemies-per-wave variable instead of a hardcoded count, consolidating enemies_remaining updates onto obj_enemy's Destroy event, moving the lives check off obj_enemy onto the player/manager, extracting player movement into its own script, and eventually retiring obj_spawner once its sole remaining job - a single start_game() call - moves onto obj_gamemanager's own Create event) - captured as raw material for the "Doing It the Doers of Stuff Way" chapter, not implemented now, with obj_spawner's own historical shrinkage across the course flagged as a good real example of incremental refactoring to walk through. Updated again to record a genuine GameMaker naming-drift fix: the drag-and-drop sound block is correctly called "Play Audio" (confirmed by Robert against his own GameMaker install), not "Play Sound"/"Audio Play Sound" as several shipped Lessons 8, 10, 11, and 12 materials had it

- corrected across the affected build scripts and reshipped, while Lesson 13's similarly-worded references were checked and confirmed to be 001 Game Creator's and Unreal's own correct, different terminology. Updated again to add a new known case to the Rule 12 reconciliation list: bullet/enemy collision ownership is opposite between GameMaker (bullet's Collision event destroys the enemy) and Godot (enemy's Hitbox Area2D destroys the bullet), found while Robert reconciled his own Godot project against the Lesson 12 code; tracked only, not unwound now by Robert's explicit decision. Updated again to record the spawn_enemies() ownership asymmetry (GameMaker: free-standing global Script resource; Godot: owned by main.gd, reached from GameManager via get_parent()) and its tangled two-way obj_spawner/main.gd non-parallel, plus Robert's request to extend the eventual Rule 12 pass to Scratch's objects/events/broadcasts as well, likely via explicit Data Dictionary cross-references rather than literal renaming. Updated again to record a Godot-only, deferred decision: whether signal connections should be written explicitly in code (.connect(), as the Keys currently show) or wired through the Editor's Signals tab (Robert's own project uses this instead), found while Robert reconciled his enemy.gd - tracked only for now, no cross-engine parallel. Updated again to record Robert's own cross-engine ownership-reassignment pattern in GameMaker and Godot (shared collision/sound logic moved to whichever object Robert personally felt should own it, while determining his own eventual refactoring preferences) as raw material for the "Doing It the Doers of Stuff Way" chapter rather than a design to keep, plus Robert's own deliberate and cross-engine-consistent bullet-spawn-offset deviation, to be noted now and removed for the book's canonical code later. Updated again, same session, to CORRECT that entry: Scratch is not an example of Robert's drift the way GameMaker and Godot are - Claude initially misread Scratch's Bullet-owns-nothing design as drift, but Scratch's own shipped Key deliberately moves collision-ownership from Bullet (Lessons 3-4) to Enemy (Lesson 5 onward) on purpose, documented in the Lesson 5 Key itself (Scratch clones have no unique ID, so only the touched Enemy clone can identify and delete itself) - a genuine third Rule-12 data point and a worthwhile Scratch-specific teaching point, not drift to reconcile away. Updated again to record Robert's confirmation, while reconciling his Scratch GameManager, that the self-reporting enemies_remaining pattern (refactoring candidate 1 from the GameMaker list) was independently built in Scratch too and is being reconciled back to canonical, plus a personal reminder to retest two Scratch boundary values (escape y and spawn y) at full screen rather than only the editor's inset window. Updated again to record Lesson 13's compliance pass - the worst drift found in any pass to date (stale obj_spawner/spawn_wave references in GameMaker and TIC-80, a Rule 4 GameMaker Visual block-choice gap, a Scratch broadcast-casing slip caught before shipping, and a real Unity/Godot timing bug where score/win-check fired after the hit-reaction delay instead of before it) plus the project's first Rule 9 Activity Diagram built in direct response to a student question, and a new GameMaker Visual node diagram showing an object gaining a brand-new event. Robert then granted standing authorization to audit and fix Lessons 14 and 15 to the same standard while away from the keyboard, to be reviewed together as completed work on his return. Updated again to record Lesson 14's compliance pass - a stale spawn_wave()-vs-spawn_enemies() naming drift affecting Scratch, Godot, and GameMaker (plus a near-miss Scratch broadcast-casing catch), a genuine Godot/GameMaker start_game() ordering bug between wave and enemies_remaining, a brief GameMaker architecture regression (enemies_remaining reset folded into spawn_enemies() itself) caught and reverted to the established two-call-site split, a new Rule 9 diagram pair, Rule 8 speaker notes, and a Rule 10 demo-engine correction (GameMaker, not Godot) - plus a genuine Rule 12 discovery that TIC-80 legitimately uses a different name (spawn_wave) than the other engines, found only because fixing the other four engines required confirming TIC-80 was not itself drifted. Two bugs accidentally introduced during the Lesson 13 fix pass were also caught and corrected while working on Lesson 14: a wrongly Title-Cased Scratch broadcast name, and a TIC-80 regression (a wrong rename plus a wrong sfx slot number) - both reshipped before

continuing. Updated again to record Lesson 15's compliance pass, completing the three-lesson Lesson 13/14/15 review Robert authorized while away from the keyboard: a stale obj_spawner reference (GameMaker, should be obj_gamemanager), a leftover reference to Lesson 12's abandoned Option 2 Unity field name (roundAudioSrc, corrected to the real waveClip), a slide-timing correction (two milestone-framing sections re-tagged from 10+10 minutes down to 5+5 to match the actual 10-minute combined block), five new Rule 8 speaker notes, and the project's first Activity Diagram built around a class-day workflow (checklist, fix, polish, present) rather than a programming concept, since this lesson introduces no new mechanic at all. All three lessons (13, 14, 15) are now fully audited, fixed, diagrammed, rebuilt, verified, and shipped, ready for Robert's review as completed work. Updated again to record Robert's "Sound Manager" proposal for the book version of Lesson 12: keep the scattered early Level 3 sound stretch goals in Lessons 2-8 (reversing the earlier strip-it-out leaning) and instead make Lesson 12 the lesson where a student's scattered sound gets explicitly refactored into one formal SoundManager object, reviving and sharpening the already-preserved Option 2 reasoning and matching the course's other extract-into-a-manager lessons (GameManager, spawn_enemies(), wave_cleared()). Updated again to record Robert's Unity Bullet.cs/Enemy.cs reconciliation - a fourth confirmed engine with the bullet/enemy collision-ownership-reassignment exploration pattern (added to the Rule 12 ToDo above), a deliberate fire-sound-on-Bullet placement choice kept as-is for the refactoring chapter, two real incidental bugs found and flagged (Enemy.cs checking CompareTag("Player") instead of "Bullet", traced to the Bullet prefab itself being tagged "Player"; and a stray, unused using UnityEditor.UI; that would break a standalone build), and a new tracked-but-not-yet-fixed continuity bug in the shipped Lesson 13 Unity materials (hit-detection wrongly described as always having lived on the Enemy's script). Updated again to record Robert's GameManager.cs reconciliation: one genuine gap (the Lesson 12 waveClip field/sound, missing entirely, added into the Stalemate-fix WaveCleared()'s else branch since his project is already past Level 1), and one legitimate kept addition (a gameStartSound played in Start(), using the sfx_start.wav asset shipped back in Lesson 6 for exactly this purpose). Updated again to record a new deferred-by-design ToDo: whether Unity's early lessons should teach [SerializeField] private over plain public fields, raised when Robert's own EnemySpawner.cs used the former while explicitly declining to request the same change in canonical - he wants this decided as its own intentional discussion, not defaulted either way, with a leaning toward keeping canonical on public fields for beginner simplicity. Updated again to record the close of the file-by-file Unity reconciliation: HUD.cs's one deviation (a GameManager.gameActive debug Text field) turned out to be a deliberate, reusable live-classroom debugging technique worth its own named sidebar - wiring internal state straight onto the visible HUD when the room can't easily see a console - not just an artifact to remove (removed from the reconciled file regardless, as debug-only). PlayerMovement.cs, the last file, had two real behavior gaps traceable to one structural choice (four separate per-key if blocks instead of canonical's single combined moving check): blinkTime could double-increment during diagonal movement, and walkingBeep.Stop() was missing entirely, meaning a looping footstep clip would never actually stop when the player went idle. Both fixed by reconciling to canonical's single-check structure; Robert's named xBoundary/yBoundary fields and Inspector-wired sr reference were both kept as legitimate improvements. Robert still needs to re-drag the Inspector references and playtest this file before it's confirmed working - not yet verified at the keyboard, unlike every other file in this reconciliation sequence. Updated again to record and correct a real error: the Bullet.cs rewrite Claude delivered had quietly dropped the audioSrc field and its Start()-time fire-sound .Play() call, directly contradicting Robert's explicit instruction to keep it and the note Claude had just written claiming it was kept - caught by Robert via the actual symptom (fire and hit sounds both missing in-game) rather than by review of the code itself. Fixed by restoring audioSrc/.Play() to Bullet.cs; separately, hitSound (moved from Enemy's old enemyHit field) needs its own fresh Inspector drag onto Bullet's prefab, since it's a new field with

no carried-over assignment. Updated again to record the actual resolution: the hit sound was never missing, just very quiet, and Robert correctly diagnosed why (PlayClipAtPoint()'s 3D distance rolloff, not fixable via the volumeScale argument as Claude first suggested) and the real fix (play it at Camera.main.transform.position, matching GameManager's existing sounds) - which turned out to be a genuine, previously undocumented bug in the shipped Lesson 4 Unity materials (marked for book-only correction per Robert's decision) and a real regression in Lesson 13's HandleHit() Coroutine rewrite (the hit sound had been dropped entirely - fixed and reshipped, including a new troubleshooting note and a new Rule-9-adjacent sidebar candidate for the rolloff gotcha itself). Robert also settled his earlier fire-sound-ownership question in the other direction from his first instinct: keep it in the canonical Player-side location after all, for cross-engine consistency, saving his own ownership argument for the refactoring chapter instead - Bullet.cs's fire-sound field was removed again, this time intentionally. The canonical Player-side fire sound itself is still pending Robert's actual fire/shoot script, since PlayerMovement.cs turned out not to contain any shooting logic. Updated again to record Player.cs - the file Robert had initially missed - closing out the Unity reconciliation sequence: it already matched canonical almost exactly (one already-tracked deviation, the bulletYOffset spawn offset, now confirmed as a fourth engine with the same personal habit), with the fire sound itself the only real gap, added as a SerializeField AudioSource with audioSrc.Play() right next to Instantiate(), exactly where canonical puts it - no Camera.main.transform.position fix needed here, since a persistent Inspector-added AudioSource defaults to 2D spatial blend and was never subject to PlayClipAtPoint()'s rolloff problem. Robert considers Unity now returned to [mostly] canonical across all seven reconciled files (Bullet, Enemy, GameManager, EnemySpawner, HUD, PlayerMovement, Player). Also added, per Robert's explicit request, a personal marker note (not course content) recording Lesson 12 as the Git checkpoint where his own experimental Unity code was deliberately stripped back out in favor of canonical, so he can find his pre-reconciliation code again later if he wants to revisit it. Add to this document as the course continues; do a dedicated consolidation pass every so often rather than continuously.